

GOVERNMENT OF ZAMBIA

STATUTORY INSTRUMENT NO. 79 OF 2013

The Electricity Act
(Laws, Volume 24, Cap. 433)

The Electricity (Grid Code) Regulations, 2013

IN EXERCISE of the powers contained in section *thirty* of the Electricity Act, the following Regulations are hereby made:

- | | |
|---|-------------------------|
| 1. These Regulations may be cited as the Electricity (Grid Code) Regulations, 2013. | Title |
| 2. In these Regulations, unless the context otherwise requires —
“ Code ” means the Electricity Grid Code set out in the Schedule; and
“undertaking” has the meaning assigned to it in the Act. | Interpretation |
| 3. The provisions of the Schedule apply to undertakings in the electricity supply industry. | Application of Schedule |

GOVERNMENT OF ZAMBIA

Statutory Instrument No. 79 of 2013

SCHEDULE

(Laws, Volume 2, Cap 133)
(Regulation 3)

THE ZAMBIAN GRID CODE

Is the power contained in section thirty of the Electricity Act, the following Regulations and hereby made.

1. These Regulations may be cited as the Electricity (Grid Code) Regulations, 2013.

2. In these Regulations, unless the context otherwise requires, "Code" means the Electricity Grid Code set out in the Schedule; and

"undertaking" has the meaning assigned to it in the Act.

3. The provisions of the Schedule apply to undertakings in the electricity supply industry.

CONTENTS

Preface

1	Scope	629
2	Objectives	629
3	Responsibility	629
1	Acronyms / Abbreviations	631
2	Definitions	632-7
CHAPTER 1 - GOVERNANCE		
1	Scope	638
2	Governance Structure	638
2.1	Energy Regulation Board (ERB)	638
2.2	Grid Code Technical Committee	638
2.2.1	Functions of the GCTC	638
2.2.2	Composition of GCTC	638
2.3	Operation of the GCTC	639
2.3.1	Schedule of Meetings	639
2.3.2	Chair	639
2.3.3	Procedures and Code of Conduct	639
2.3.4	Alternate Representation	639
2.3.5	Quorum	639
2.3.6	Record Keeping	640
2.3.7	Funding	640
2.3.8	Transmission Planning Coordination Oversight	640
2.4	The Grid Code Secretariat	640
3	Registration of Grid Code Participants	640
3.1	Registration and De-registration	640
3.2	Obligation of participants	640
4	Amendment Procedure	641
5	Exemption Procedure	641
5.1	Unforeseen Circumstances	642
6	Complaint Reporting, Dispute Resolution and Appeal Mechanism	643
6.1	Complaint Reporting	643
6.1.1	Complaints about the operations of the Secretariat or GCTC	643
6.1.2	Complaints between customers and service providers	643
6.1.3	Non-Conformance Report	643
6.2	Disputes	644
8.2.1	Submission of a Dispute to ERB	644
7	Grid Code Compliance Audits	645

CHAPTER 2—NETWORK

1	Application for Transmission Connections	646
2	Connection Conditions	646
2.1	Generator connection conditions	646-8
2.1.1	Protection (GCR 1)	649
2.1.2	Excitation system requirements (GCR2)	650
2.1.3	Reactive capabilities (GCR3)	651
2.1.4	Governing (GCR4)	651
2.1.5	Black starting (GCR5)	652
2.1.6	Emergency unit capabilities (GCR6)	652
2.1.7	Facility for independent generator action. (GCR7)	652
2.1.8	Testing and compliance monitoring	653
2.1.9	Non-compliance suspected by the SO	653
2.1.10	Unit modifications	654
2.1.11	Equipment requirements	654
2.2	Distributors and end-use customers	654
2.2.1	Power factor	654
2.2.2	Protection	655
2.2.3	Fault levels	655
2.2.4	Network Performance	655
2.2.5	Equipment requirements	655
2.2.6	Additional Reinforcement	655
3	TNSP Technical Design Requirements	656
3.1	Equipment design standards	656
3.2	Clearances	656
3.3	CT and VT ratios and cores	656
3.4	Standard busbar arrangements and security criteria	656
3.5	Motorised Isolators	656
3.6	Earthing and Surge Protection	656
3.6.1	Earthing isolators	657
3.7	Tele-control	657
3.8	Substation drawings	657
3.9	Recorders	657
3.10	Fault Levels	658
3.11	The TNSP's delivered QOS	658
4	Service Provider Protection Requirements	658
4.1	Equipment protection requirements	658
4.1.1	Feeder protection: above 132kV	658
4.1.2	Feeder protection: 66kV and below, at TNSP substations	659
4.1.3	Tele-protection requirements	659
4.1.4	Transformer and reactor protection	659

4.1.6	Transmission bus coupler and bus section protection	660
4.1.7	Transmission shunt capacitor protection	660
4.1.8	Over-voltage protection	660
4.1.9	Ancillary protection functions	660
4.2	System protection requirements	661
4.2.1	Under-frequency load shedding	661
4.2.2	Out-of-step tripping	661
4.2.3	Under-voltage load shedding	661
4.3	Protection system performance monitoring	661
5	Nomenclature	662
6	TS Planning and Development	662
6.1	Planning process	662
6.2	Forecasting the demand	662
6.3	Technical limits and targets for planning purposes	662
6.3.1	Voltage limits and targets	662
6.3.2	Other targets for planning purposes	662
6.3.3	Reliability criteria for planning purposes	663
6.3.4	Contingency criteria for planning purposes	663
6.3.5	Integration of Power Stations	664
6.3.6	Least economic cost criteria	664
6.3.7	Cost reduction investments	665
6.3.8	Statutory or strategic investments	665
6.4	Development investigation reports	665
6.5	Transmission investment plan	666
6.6	Mitigation of network constraints	666
6.7	Interfacing between participants and TNS Ps	666
6.8	Special end use customer requirements for increased reliability	666
7	Network Maintenance	666

CHAPTER 3 - METERING

1	Objective	667
2	Scope	667
3	General Provisions	667
4	Responsibility for Metering Installations	667
5	Metering Installation Requirements	668
6	Data Validation	668
7	Meter Verification	669
8	Metering Database	669
9	Metering Database Inaccuracies	669
10	Access to Metering Data	669
11	Confidentiality	669

CHAPTER 4 - SYSTEM OPERATION

1	Objective	670
2	Scope	670
3	Responsibility	670
4	So Responsibilities	670
4.1	System reliability, safety and security	670
4.2	Operational measures	671
5	Scheduling of Generation and Ancillary Services	671
6	Ancillary Services	671
7	Operational Authority	671
8	Operating Procedures	672
9	Operational Liaison	672
10	Emergency And Contingency Planning	672
11	System Frequency and Ace Control Under Abnormal Frequency or Imbalance Conditions	673
11.1	Description of normal frequency or balancing conditions	673
11.2	Operation during abnormal conditions	673
12	Independent Action by Participants	673
13	Voltage Control	674
14	Fault Reporting, Analysis and Incident Investigation	674
14.1	Generators, TNSPs, Distributors and End-use Customers	674
14.2	System Operator	674
14.3	Regional Operator	674
14.4	Root Cause and Forensic Analysis	675
15	Commissioning	675
16	Maintenance Coordination/Outage Planning	676
16.1	Outage Management	676
16.1.1	Yearly Planned Maintenance Schedule	676
16.1.2	Yearly Unplanned Outages	676
16.1.3	Effecting of Outages	676
16.2	Emergency Outage	676
16.3	Long Term Maintenance Planning for Generators	676
16.4	Refusal/cancellation of outages	677
17	Communication Of System Conditions, Operational Information and IPS Performance	677
18	Tele-Control	677

CHAPTER 6 - INFORMATION EXCHANGE

1	Objective	678
2	Scope	678
3	Precedence	678
4	Information Exchange Interface	678
5	System Planning Information	678
5.1	Objective	678
5.2	Information required by TNSPs	678
5.3	Information required by Customers	679
5.4	Information required by Generators	679
5.5	Information required by the SO	679
6	Operational Information	679
6.1	Pre-commissioning studies	679
6.2	General information requirements	679
6.3	Commissioning and notification	680
6.4	Inter control centre communication	680
6.4.1	Normal Conditions	680
6.4.2	Abnormal Conditions	681-2
7	Post-Dispatch Information	683
7.1	System information	683
7.2	Generation information settlement	683
7.3	Additional post-dispatch information	683
7.4	Half-hourly demand metering data	683
8	File Transfers	683
9	Performance Data	683
9.1	Generator performance data	683
9.2	Distributor and end-use customers performance	683
9.3	TNSP and SO performance	684
9.4	System Operation Performance Information	684
9.4.1	Daily	684
9.4.2	Monthly	684
9.4.3	Annually	684
10	Confidentiality of Information	684

CHAPTER 7 - APPENDICES

Appendix 1: Amendment Request Form	685
Appendix 2: Exemption Request Form	686
Appendix 3: Amendment and Exemption Request Log	687
Appendix 4: Register of Approval Exemptions	688
Appendix 5: Log of Disputes between Participants	689
Appendix 6: Standard Application Form for Transmission Connection	690-3
Appendix 7: Surveying, monitoring and testing for generators	694-702
Appendix 8: Transmission drawings symbol set and layout conventions	703
Appendix 9: Technical Voltage Limits	704
Appendix 10: Typical Load Profile	705
Appendix 11: Load Duration Curve	706
Appendix 12: Operation during abnormal conditions	707
Appendix 13: Format for Preliminary Incident Report	708
Appendix 14: Distributor and End-use Customer data	709-14
Appendix 15: Generator Planning Data	715-19
Appendix 16: Generator Maintenance Data	720
Appendix 17: Information requirements for upgrading of existing connections	721
Appendix 18: Generator HV Yard Information	722
Appendix 19: Operational Data	723-32
Appendix 20: Post-dispatch Information	733
Appendix 21: Backup Files	734
Appendix 22: Generator Performance Data	735-37
Appendix 23: Reporting format for periodic testing of under-frequency load shedding relays	738
Appendix 24: Performance indicators for TNSPs and SO	739
Appendix 25: Confidentiality of Information	740-41
Appendix 26: Planning Schedules	742

SCOPE OF CODE

This Code applies to all undertakings.

The elements of the industry structure for which this Code applies are as follows—

- (a) an SO and the national Transmission Network Service-Provider (TNSP);
- (b) a Regional Operator and independent TNSP, roles that are currently with CEC;
- (c) a generation sector consisting of ZESCO-owned generators and independent generators;
- (d) a distribution sector;
- (e) end-use customers, buying directly from a generator or being supplied via a supplier; and
- (f) international trading via the interconnectors with other countries, and in line with the SAPP rules.

OBJECTIVES

This Code shall regulate the reciprocal obligations of industry participants on the use of the TS and operation of the IPS.

The Code provides for the following—

- (a) minimum technical requirements for customers connecting to the TS;
- (b) minimum technical requirements for service-providers;
- (c) that SO obligations defined to ensure the integrity of the IPS;
- (d) that obligations of participants defined for the safe and efficient operation of the TS; and
- (e) that relevant information is made available to and by the participants.

This Code shall —

- (i) ensure that investments are made within the requirements of the Code; and
- (ii) provide access, on agreed standard terms, to all parties wishing to connect to or use the TS.

This Code shall apply the principle of non-discrimination through the provision of consistent and transparent principles, criteria and procedures.

RESPONSIBILITY

The ERB has the responsibility of ensuring compliance with this Code for the benefit of the electricity industry and consumers.

The Grid Code Secretariat is responsible for implementation, maintenance and revision control of this Code and shall ensure that all recipients of this Code have the latest revision of the Code.

The Grid Code Secretariat shall ensure that it has the latest copies of relevant standards that have been quoted in this Code.

I. ACRONYMS / ABBREVIATIONS

Note: Standard SI symbols and abbreviations are used throughout the Code.

AAICG	Annual Average Incremental Cost of Generation
AC:	Alternating Current
ACE:	Area Control Error
AGC:	Automatic Generation Control
ARC:	Auto Re-close
AVR:	Automatic Voltage Regulator
A-U/F-LS:	Automatic Under-Frequency Load Shedding
B/U:	Back-Up
BSA:	Bulk Supply Agreement
CAPEX:	Capital Expenditure
CBM:	Condition Based Maintenance
CEC:	Copperbelt Energy Corporation
CT:	Current Transformer
DC:	Direct Current
DCF:	Discounted Cash Flow
DCS:	Distributed Control System
DEF:	Directional Earth Fault
EENS:	Expected Energy Not Served
E/F:	Earth Fault
ERB:	Energy Regulation Board
ERLF:	External Reliability Loss Factor
ESI:	Electricity Supply Industry
FACTS:	Flexible AC Transmission System
F/L:	Fault Level
GAV:	Gross Asset Value
GCR:	Grid Code Requirement
GCTC:	Grid Code Technical Committee
GPS	Global Positioning System
HV:	High Voltage
HVDC:	High Voltage Direct Current
Hz:	Hertz
IDMT:	Inverse Definite Minimum Time
IEC:	International Electro-technical Commission
IEEE:	Institute of Electrical and Electronic Engineers
IPS:	Interconnected Power System
LV:	Low Voltage
MCR:	Maximum Continuous Rating
MV:	Medium Voltage
MVA:	Mega-Volt-Ampere
MVA _r :	Mega-Volt-ampere reactive

MW:	Mega-Watt
NAV	Net Asset Value
NCR:	Non-Conformance Report
NERC:	North American Electricity Reliability Council
NPV:	Net Present Value
OEM:	Original Equipment Manufacturer
O&M:	Operation and Maintenance
OST:	Out-of-Step Tripping
O/C:	Over-Current
PCC:	Point of Common Coupling
PCLF:	Planned Capability Loss Factor
PCS:	Process Control System
PPA:	Power Purchase Agreement
Protection	Grid Protection
PSA:	Power Supply Agreement
PSB:	Power Swing Blocking
p.u.:	per unit
QOS:	Quality of Supply
RTU:	Remote Terminal Unit
REA:	Rural Electrification Authority
	No. 20 of 2003
RO:	Regional Operator
SAPP:	Southern African Power Pool
SCADA:	Supervisory Control and Data Acquisition
SCS:	Substation Control System
SFR:	Start-Up Failure Rate
SO:	System Operator
SSR:	Sub-synchronous Resonance
STATCON:	Static Condenser
SVC:	Static VAR Compensator
TCSC:	Thyristor Controlled Series Capacitor
TNSP:	Transmission Network Service-Provider
TS:	Transmission System
Tx:	Transformer
UCF:	Unit Capability Factor
UCLF:	Unplanned Capability Loss Factor
Um, Umax:	Maximum rated voltage
Un:	Nominal voltage
UPC:	Unified Power Controller
VAR:	Voltage Ampere Reactive
VT:	Voltage Transformer
ZS:	Zambian Standard

2 DEFINITIONS

In this Code, unless the context otherwise requires—

- “academia” means research and teaching staff of institutions of learning offering post-secondary education;
- “ancillary services” means services supplied to the TS by generators, distributors or end-use customers necessary for the reliable and secure transport of power from generators to distributors and customers in order to maintain the short-term reliability of the IPS and includes—
 - (a) operating reserves;
 - (b) black start;
 - (c) reactive power supply; and
 - (d) regulation and load following;
- “area control error” means the mismatch between the instantaneous demand and supply of a control area, which combines the frequency error and the tie line schedule error;
- “automatic generation control” means the regulation of the power output of electric generators within a prescribed area in response to change in system frequency, tie-line loading, or the relation of these to each other, so as to maintain the scheduled system frequency or the established interchange with other areas within predetermined limits or both
- “black start” means the provision of generating equipment that, following a total system collapse (black out), is able to—
 - (a) start without an outside electrical supply; and
 - (b) energise a defined portion of the TS so that it can act as a start-up supply for other capacity to be synchronised as part of a process of re-energising the TS;
- “budget quotes” means a provisional invoice stating connection conditions, inclusive of financial terms;
- “busbar” means an electrical conduit at a substation where lines, transformers and other equipment are connected;
- “connection (connected) to the TS” means physical connection of customer equipment to the TS either directly or through a dedicated transformer provided by the TNSP;
- “constrained generation” means the difference between the energy scheduled under normal operating conditions and the energy scheduled under constrained conditions at a point of connection;
- “control area” means an electrical system with borders defined by points of interconnection and capable of maintaining continuous balance between the generation under its control, the consumption of electricity in the area and the scheduled interchanges with other control areas;
- “control centre” means an entity responsible for the operational control of the entire or specified electricity network assets;
- “customer” means an entity that contracts directly with the service-provider for the provision of transmission services. These include generators, distributors, end-use customers and suppliers;
- “day” means a period of 24 consecutive hours commencing at 00:00;
- “demand side managed load” means load that may be reduced (or increased) in response to a signal from the System Operator and it includes interruptible load, ripple controlled residential boilers and dual fuel boilers, but excludes under-frequency customer load shedding;

“demarcation point” means the point at which there is a change over in ownership from service-provider to customer.

(a) TNSP—Power Stations

The HV bushing stems of the unit step-up transformers. The protection circuits of the units and step-up transformers shall belong to the generator; all other protection circuits and equipment in the HV yard shall belong to the TNSP; and

(b) TNSP—Distributors, international tie lines and end-use customers.

The demarcation point shall be at the point where ownership changes hands;

“dependability” means the probability of not failing to operate under given conditions for a given time interval [IEC 50 – 448];

“distribution system” means an electricity network consisting of assets operated exclusively at a nominal voltage as defined in ZS387 – 1: Power Quality and Reliability Standard;

“distributor” means a person or entity that owns, operates or distributes electricity through a distribution system;

“electricity” means electrical produced by physical sources of energy such as hydro-power, wind power, solar power, petroleum, coal, biomass, nuclear energy and any other source;

“electricity supply industry” means the industry in Zambia involved in generation, transmission, distribution and supply of electrical energy.

“emergency” means a situation where there is unplanned loss of generation, transmission or demand facilities that jeopardizes the ability to meet system requirements;

“emergency level generation” means extra capacity, achieved without significant additional cost, from generating units over and above their MCRs that can be supplied up to one hour without risk of damage to the plant;

“emergency outage” means an outage which occurs when plant has to be taken out of service immediately so that repairs can immediately be effected to prevent further damage or loss;

“end-use customer” means a consumer of electricity who, or which, is connected to the TS;

“expert team” means a team of experts established by the GCTC;

“firm quote” means a form of contract negotiated and signed with a customer stating final connection conditions, including financial terms;

“firm supply” means a supply that enjoys a level of reliability as specified in the Network Chapter;

- " flicker " means a cyclic voltage fluctuation, normally between 0.1Hz and 10Hz, that causes optical stress to humans;
- " force majeure " means any of the following events that prevents any participant from executing the participants obligations laid down in the Code:
- (a) any overwhelming occurrence of nature that could not reasonably have been foreseen or guarded against;
 - (b) war, blockade, foreign hostile acts, civil war, rebellion, revolution, insurrection, sabotage or strikes or other similar stoppages of work by employees that are not caused by unreasonable actions on the part of the participants; and
 - (c) any other cause beyond the control of the participants, which the affected participants may by agreements regard as *force majeure*;
- " forced outage " means an outage that is not a planned outage;
- " frequency " means the number of oscillations per second on the AC waveform;
- " generator " means an entity operating a generating unit or power station;
- " governing " means a mode of operation where any change in system frequency beyond the allowable frequency dead band will have an immediate effect on the unit output according to the governor droop characteristic;
- " Grid Code Secretariat " means the entity responsible for the administrative functions as defined in the Governance Chapter;
- " Grid Service Charge " means the operating costs of the SO to ensure safe and reliable operation of the IPS. This includes funding of the Secretariat;
- " Gross Asset Value " means the total cost of an asset to the TNSP representing the capital cost, cost of purchase, freight and insurance, and installation;
- " Incident Report " means a formal communication of an occurrence between parties that results in disagreements;
- " individual customer charge " means connection costs that are peculiar to a particular customer;
- " information " means a type of knowledge represented by some data, which can be exchanged, stored or processed electronically or otherwise;
- " information owner " means the party to whose system or installation the information pertains;
- " Interconnected Power System " means the electrical network that has a measurable influence at transmission level, consisting of –
- (a) TS;
 - (b) Assets connected to the TS and belonging to a TNSP;
 - (c) Power stations with a capacity of more than 100 KVA and networks linking such power stations to the TS;
 - (d) International inter-connectors; and
 - (e) the Control area for which the SO is responsible;

- "interruptible load" means consumer load or a combination of consumer loads that can be contractually interrupted or reduced, without notice, on instruction by the System Operator;
- "islanding" means the capability of generating units to settle down at nominal speed, supplying own auxiliary load after separation from the grid, at up to full load pre-trip conditions;
- "load curtailment" means load reduction by customers who are willing and able to reduce their use of power within 1 hour on instructions from the SO;
- "load following" means the provision of generation and load response capability, including capacity, energy, and manoeuvrability, that is dispatched by the SO to match power generation and load demand within a scheduling period;
- "load reduction" means the ability to reduce customer demand by load curtailment and load shedding;
- "losses" means the technical or resistive energy losses incurred on the TS;
- "maximum continuous rating" means the capacity that a generating unit is rated to produce continuously under normal conditions;
- "metering" means equipment used in measuring supply parameters installed at supply points;
- "metering installation" means an installation that comprises an energy meter that is interrogated and has a communication link;
- "month" means a calendar month comprising a period commencing at 00:00 hours on the first day of that month;
- "NERC A1 and A2 criteria" means the criteria of the NERC that an ACE must pass through zero within 10 minutes of its previous zero (A1) and that the average ACE in each 10-minute period must be within a specified limit (A2);
- "net asset value" means the value an asset has after deducting all the accumulated depreciation for that asset;
- "outage" means an interruption of the flow of power to a point of supply;
- "outage request" means a written notice from a distributor, generator, end-use customer or TNSP for plant to be taken out of service for planned maintenance, repairs, auditing, emergency repairs, construction, refurbishment, inspection, testing or to provide safety clearance for other activities such as servitude clearance, line crossings and underpasses;
- "participant" means—
- (a) a generator;
 - (b) an end-use customer;
 - (c) a distributor;
 - (d) a supplier;
 - (e) a transmission network service-provider;
 - (f) an embedded generator;
 - (g) the System Operator; or
 - (h) a Regional Operator;

- “party” means a current or future participant;
- “planned outage” means a scheduled outage of equipment that is on a yearly maintenance schedule for any year.
- “plant” means any generation unit, or switching or transformation equipment, connected to the TS;
- “point of common coupling” means the electrical node, normally a busbar, in a transmission substation where different feeds to customers are connected together for the first time;
- “point of connection” means the electrical node in a transmission substation where a customer’s assets are physically connected to the TNSP’s assets;
- “point of supply” means a transmission substation where energy can be supplied to a distributor or end-use customer;
- “power” means the rate of generating, transferring or using energy;
- “power station” means one or more units at the same physical location;
- “primary substation equipment” means high voltage equipment installed at a substation;
- “prudent utility practice” means the standards, practices, methods and procedures that conform to safety and legal requirements which are attained by exercising a degree of skill, diligence, and foresight which would reasonably and ordinarily be expected from skilled and experienced operatives engaged in the same type of undertaking under the same or similar circumstances;
- “quote” means an invoice stating connection conditions, inclusive of financial terms;
- “regional operator” means an entity, subordinate to the SO and independent from other market participants responsible for short-term reliability of the IPS, which is in charge of controlling and operating part of the TS in real time;
- “regulating reserve” means generation capacity or demand side managed load available to respond within 10 minutes. This reserve category reserves capacity as part of the regulation ancillary service. The purpose of this is to allow for enough capacity to control the frequency and control area tie-lines power within acceptable limits in real time.
- “regulation” means the provision of generation and load response capability, including capacity, energy and manoeuvrability, that responds to automatic control signals issued by the System Operator;
- “scheduling” means a process to determine which unit or equipment will be in operation and at what loading;
- “security” means the probability of not having an unwanted operation;
- “sensitivity analysis” means an exercise to determine the effect of changes in given parameters on an outcome;
- “service-provider” means a TNSP or the System Operator;
- “stakeholders” means an entity having an interest in the ESI;

“substation” means a site at which switching and/or transformation equipment is installed;

“system frequency” means the frequency of the fundamental AC voltage as measured at selected points by the System Operator;

“system minutes” means a performance indicator for interruptions, defined in the following equation:

$$S_t = \frac{EI_t * 60}{PD_t}$$

Where—

(a) S = system minutes interrupted

(b) EI = energy interrupted (MWh)

(c) PD = peak demand

(d) t = period under review

“system operator” means an entity responsible for short-term reliability of the IPS, which is in charge of controlling and operating the TS and dispatching generation or balancing the supply and demand in real time;

“transmission” means the conveyance of electricity through the TS;

“transmission equipment” means a cable, overhead line, transformer, switchgear and equipment for ancillary services used for transmission purposes;

“transmission network service-provider (TNSP)” means an entity that is licensed to own and maintain transmission equipment;

“transmission system” means part of the IPS which supplies power in bulk from power stations to distributors and other customers and includes—

(a) all transmission lines and substation equipment on the IPS where the nominal voltage is as defined in the ZS387 – 1: Power Quality and Reliability Standard; and

(b) all associated equipment at TNSP substations belonging to the TNSP;

“undertaking” means any undertaking for the generation, transmission, distribution or supply of electricity, and includes such as undertaking that generates, transmits, distributes or supplies electricity solely or mainly in the interest of a group of associated companies for the purpose of the business of those companies, whether or not any excess electricity is supplied to any other consumer who is not part of the group.

“unit” means a device used to produce electrical energy;

“unplanned outage” has the same meaning as forced outage;

“use of system charge” means a charge applied by a TNSP to enable the TNSP recover its cost of installation and maintaining its network;

“Zambian Grid Code” means this Code; and

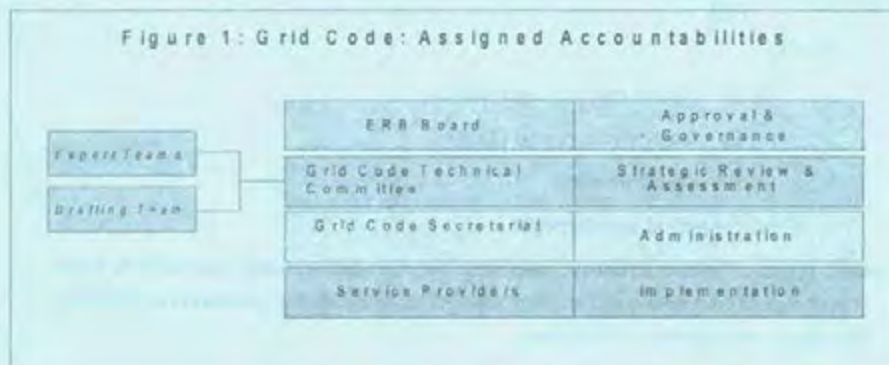
“ZESCO” means the power company incorporated in the Republic of Zambia and having its registered office at stand number 6949, Great East Road, P O Box 33304, Lusaka-10101-Zambia.

CHAPTER 1 – GOVERNANCE

1 SCOPE

This Chapter shall apply to all the aspects of this Code. In the event of any conflict with existing PPAs, BSAs and such other similar agreements, this Code shall take precedence, save where the affected parties have been granted exemptions as provided for under this Code.

2 GOVERNANCE STRUCTURE



1.

2.1 Energy Regulation Board (ERB)

The ERB, established by the Energy Regulation Act, Cap 436 of the Laws of Zambia, shall ensure the implementation of this Code.

2.2 Grid Code Technical Committee

There shall be a Grid Code Technical Committee (GCTC) constituted every two years by the ERB.

2.2.1 Functions of the GCTC—The GCTC shall have the following functions—

- (a) to review and make recommendations regarding proposals to amend is Code;
- (b) to make recommendations regarding exemptions to specific provisions of this Code;
- (c) to appoint technical experts on specific matters related to this Code; attend to the resolution of Non-Conformance Report incidences;
- (d) issue guidance in relation to this Code and its implementation, performance and interpretation when requested to do so by the participants; and
- (e) consider changes to this Code arising from unforeseen circumstances.

2.2.2 Composition of GCTC—The GCTC shall be composed of representatives of all stakeholders of the ESI in Zambia as follows—

- (a) one member representing the SO;
- (b) one member representing the RO or ROs;
- (c) two members representing TNSPs;

- (d) two members representing generators;
- (e) one member representing Distributors;
- (f) one member representing the Electrical Engineering Academia;
- (g) one member representing the Engineering Institution of Zambia Council;
- (h) one member representing the Ministry responsible for energy;
- (i) two members representing End-use customers; and
- (j) one member representing REA.

The ERB shall make available publicly the latest list of GCTC members within 14 days of any change.

2.3 *Operations of the GCTC*

2.3.1 *Schedule of Meetings*

The GCTC shall meet at least once every three months. The calendar of meetings shall be set at the first meeting, and the Grid Code Secretariat shall be responsible for sending notices of such meetings.

2.3.2 *Election of Chairperson and Vice-Chairperson*

The members of the GCTC shall elect a Chairperson and a Vice-Chairperson from amongst themselves at the beginning of every year. The positions of Chairperson and Vice-Chairperson shall be elective on an annual basis. In the event that both the Chairperson and the Vice-Chairperson are unable to attend a meeting, the members present at the meeting shall elect, from among the members, an alternative chairperson for the duration of that meeting.

Representatives of the ERB, SO and Ministry responsible for energy shall not qualify for election to the position of Chairperson.

2.3.3 *Procedures and Code of Conduct*

The GCTC shall determine its own meeting procedures and code of conduct for its members. The procedures shall set out the timelines for the processing of amendments by the GCTC.

2.3.4 *Alternate Representation*

All members of the GCTC will nominate alternate representatives in writing from among their representative groups, to attend meetings at which they are unable to be present. A register of alternate members shall be maintained by the secretariat.

Alternate members will have voting rights.

2.3.5 *Quorum*

A quorum shall consist of seven (7) members of the GCTC. Decisions by the GCTC shall be taken by consensus means of a simple majority vote of the duly constituted GCTC except for amendments to the Governance Chapter, which shall require a two-third majority vote.

2.3.6 Record Keeping

Proceedings of the GCTC meetings shall be recorded during all meetings, and shall be kept by the Grid Code Secretariat. The SO shall appoint the secretary for the GCTC meetings.

2.3.7 Funding

The Grid Code Secretariat shall fund the administrative activities of the GCTC, on a cost-recovery basis. At the beginning of every year, the GCTC shall prepare an operating budget proposal for the following year and submit it to the Grid Code Secretariat for implementation. The secretariat shall fund the operations of the GCTC and its associated committees up to the ERB approved budget and recover the costs from Grid Service Charges.

2.3.8 Transmission Planning Coordination Oversight

The GCTC shall have the duty of overseeing the coordination of transmission network planning between the TNSPs that is required under the Network Chapter section 8.7.

2.4 The Grid Code Secretariat

The SO shall serve as the Grid Code Secretariat.

The Grid Code Secretariat shall perform the following functions—

- (a) co-ordinate the activities of the GCTC;
- (b) keep a register of licensed undertakings;
- (c) submit amendments and exemptions to the ERB following review by the GCTC; and
- (d) manage the documentation relating to this Code and disseminate information to parties.

3 REGISTRATION OF GRID CODE PARTICIPANTS

3.1 Registration and De-registration

The Grid Code Secretariat shall be responsible for making entries in the register of participants upon receipt of notification from the ERB of licensed undertakings. Participants shall be registered in different categories: Generator, Embedded generator, Distributor, End-use customer and TNSPs.

An undertaking shall not have access to the TS before obtaining a licence from the ERB.

Service-providers shall ensure that distributors and end-use customers are registered as participants before entering into a contract for services with such customers.

A participant whose licence has been withdrawn by the ERB ceases to be a participant.

3.2 Obligation of participants

The provisions of this Code are binding on participants and the participants shall comply with the said provisions unless exempted.

4 AMENDMENT PROCEDURE

A party, a member of the GCTC, or the ERB, may request amendments to this Code in the form set out in Appendix 1 of this Code.

A participant or member of the GCTC proposing an amendment of this Code shall record the proposed amendments on the prescribed form.

All proposed and approved amendments shall be logged on the Request Log as shown in Appendix 3.

The Grid Code Secretariat shall forward the proposed amendment to the GCTC for the review process.

If the proposed amendment requires expert opinion, the GCTC shall determine the necessity for constituting a team of experts. The GCTC shall constitute a team of experts for the purpose of allowing expert opinion to be obtained regarding a particular proposed amendment. The team of experts so constituted shall report back to the GCTC within a specific period.

The proposer of an amendment shall attend the GCTC sessions and may attend the expert team sessions and shall be allowed to make presentations if necessary.

The GCTC or the expert team may refine the amendment or make alternative amendments to achieve the same purpose. The proposer of the amendment may or may not agree that any refined or alternative amendment achieves the proposer's original purpose.

Once the GCTC has reviewed submissions, it shall make a formal submission to the ERB through the Secretariat on all proposed amendments to the Code other than those amendments proposed by ERB or the representative of the Ministry's responsible for energy or the GCTC, where the GCTC has the right to reject them. The GCTC shall give the decision it reached on each proposal, state the impact of the proposals on the Code, and also provide the ERB with divergent views on such proposals. The GCTC shall also describe its areas of discussion, findings and any recommendations made. In particular the ERB shall be informed whether the proposer continues to support the refined or alternative amendment.

The GCTC submission to the ERB shall include the text of the proposed amended Code. The GCTC may convene a Drafting Team if the documentation of the draft amendment so demands.

The ERB shall inform the Grid Code Secretariat of the ERB's decisions regarding the proposed amendments and the Secretariat shall communicate decisions to all participants.

The ERB shall notify the Minister, in writing, of the proposed amendments approved by the ERB and the Minister may issue a statutory instrument amending the Code accordingly.

5 EXEMPTION PROCEDURE

Applications for exemption from complying with any provision of this Code may be made in the form set out in Appendix 2 of this Code for any of the following reasons—

- (a) to provide for existing equipment that has not been designed with consideration of the provisions of the Code;
- (b) to facilitate transition through interim arrangements;

- (c) to facilitate temporary conditions necessitating exemption; or
- (d) contractual obligations such as PPAs entered into prior to the coming into effect of this Code.

All proposed and approved exemptions shall be logged on the Request Log as shown in Appendix 3.

If a request requires expert opinion, the GCTC shall determine the necessity for constituting a team of experts. The GCTC shall constitute a team of experts for the purpose of allowing expert opinion to be obtained regarding a particular application, which shall report back to the GCTC within a specific period.

An applicant for exemption may attend the GCTC sessions as well as the expert team sessions and will be allowed to make written or oral presentations.

Once the GCTC has reviewed submissions, it shall make a formal submission to the ERB through the Secretariat on the application for exemption. The GCTC shall give the decision it reached on each application and also provide the ERB with divergent views on such applications.

The ERB shall inform the Secretariat of the decision reached in respect of each application and Secretariat shall communicate the decision to the respective applicants and, except where an appeal has been lodged with the Minister, participants.

Full or partial exemption from complying with certain provisions of the Code may be granted to a participant that has applied for that exemption.

The Grid Code Secretariat shall maintain a register in which shall be recorded the names of the participants who have been granted exemption under this Code, the nature of the exemption so granted and its duration. The register shall be in the form shown as appendix 4.

An applicant who is dissatisfied by the decision of the ERB not to grant the application for exemption may appeal to the Minister within thirty days from the date of being notified about the decision.

5.1 *Unforeseen Circumstances*

It shall be the responsibility of the SO where unforeseen circumstances arise to the extent reasonably practicable under the prevailing circumstances to promptly consult all affected participants in an effort to reach an agreement as to the appropriate or necessary action to be taken. If an agreement is reached between the relevant participants and the SO, it shall be the responsibility of the SO to promptly refer the matter, including the agreement, to the GCTC for review and to make the necessary recommendations to the ERB.

In the event of an agreement not being reached between the participants and the SO, the SO shall decide on the next course of action to be taken if the security of the Grid is in question. In such cases, all participants shall comply with all instructions issued by the SO to the extent that such instructions are consistent with the technical characteristics of the relevant participant's system under the Code. The GCTC and ERB shall superintend over the SO to ensure that no unilateral measures or actions are taken that will prejudice any participant.

For the avoidance of doubt where unforeseen circumstances occur in real time the SO shall take whatever actions it considers necessary to secure the system. Subsequently the SO shall submit details of the problem and its actions to the GCTC.

6 COMPLAINT REPORTING, DISPUTE RESOLUTION AND APPEAL MECHANISM

The procedure for handling disputes arising from the implementation of the Code shall be as follows.

6.1 *Complaint Reporting*

6.1.1 Complaints about the operations of the Secretariat or GCTC

Any complaint regarding the operations of the Secretariat or the GCTC shall firstly be addressed in writing to the Secretariat. The GCTC shall attend to such complaints at or before the next scheduled meeting. If the complaint is not resolved, the matter shall be referred to the ERB as a dispute and shall follow the procedure described in Section 6.2.

6.1.2 Complaints between customers and service providers

Complaints arising between Customers and Service Providers shall be handled in accordance with section 6.1.3 and section 6.1.4.

A customer may issue an incident report to a service-provider on becoming aware of an occurrence. The service-provider shall provide a reason for the occurrence, what has been done to address it, and, if appropriate, indicate what action it shall take to avoid such an incident in the future.

A service-provider may also issue an incident report to a customer, where the customer does not comply with necessary requirements. The customer shall provide the service-provider with reasons for the occurrence and, where appropriate, indicate the measures that will be taken to address the problem.

Service-providers shall keep a log of all incident reports received and a log of all incident reports sent to customers.

Incident reports are operational in nature and generally require action only by technical and customer-relations staff.

6.1.3 Non-Conformance Report

NCRs shall be generated to indicate problems that require GCTC's intervention.

A customer under any of the following conditions may issue an NCR after an incident report—

- (a) the service-provider fails to provide the appropriate feedback to the incident report;
 - (b) the service-provider wilfully misrepresents the facts concerning the incident;
 - (c) the service-provider fails to implement the agreed preventative actions within the timeframe agreed by the parties;
 - (d) the number of incident reports becomes excessive in relation to historical performance;
- or
- (e) the actions arising from an ERB mediation/arbitration process are not adhered to.

A service provider under any of the following conditions may issue an NCR after an incident report—

- (a) the customer fails to provide the appropriate feedback;
- (b) the customer wilfully misrepresents the facts concerning an incident;
- (c) the customer fails to implement the agreed preventative actions within the time frame agreed by the parties;

(d) the number of incident reports becomes excessive in relation to historical performance; or

(e) the actions arising from an ERB mediation/arbitration process are not adhered to.

In the case where the parties agree and assign responsibilities outlined in the NCR, both parties, within an agreed time frame, shall implement remedial action. In the event of parties failing to agree within a set time frame, the provisions of section 6.2 may be invoked.

Service-providers shall report annually to the ERB on the following aspects of the procedure—

(a) the number of NCRs for each customer category; and

(b) the number of closed-out NCRs for each customer category.

Upon a report or suspicion of non-compliance the ERB may seek to—

(a) resolve the issue through negotiation;

(b) take action in terms of the procedures for handling licensing contraventions;

(c) advise the parties to consider an application for amendment; or

(d) advise the parties to consider an application for exemption.

6.2 Disputes

Disputes are unresolved complaints between parties that require intervention. A dispute may be declared when an NCR cannot be closed out in the timeframe agreed by the parties. At this stage the complaint shall be referred to the ERB.

6.2.1 Submission of a Dispute to ERB

Any party may submit a dispute to the ERB provided the required process of section 6.1 has been followed.

When a dispute is raised with the ERB, participants shall provide the following information—

(a) the full history of relevant incident reports;

(b) the detailed NCR and accompanying information that gave rise to the dispute; and

(c) a written report from each participant detailing the reason for not being able to close out the NCR.

Disputes received by the ERB shall be recorded in accordance with Appendix 5. The following shall be considered by the ERB in dealing with the dispute—

(a) the effectiveness of the incident resolution management system, that is to say, the manner in which the problem was addressed at each stage before a dispute was declared;

(b) how thoroughly the problem has been studied by both participants;

(c) what action has already been taken;

(d) the actual cost, environmental, or other impact on the parties;

(e) whether the complaint is reasonable; or

(f) whether a timeframe cannot be agreed for closing out an NCR.

The ERB may act as a mediator upon agreement of the parties. Where mediation fails the parties may refer the matter for arbitration in accordance with the Arbitration Act No. 19 of 2000.

The ERB shall continue to develop a database of precedents based on disputes resolved. These precedents shall be used in rulings on complaints or disputes. However, precedents set by any other parties in attempting to resolve an NCR shall not be binding on the ERB.

The ERB shall consider the following, among other things, during the dispute resolution process:

- (a) existing and historical performance trends;
- (b) reference standards;
- (c) the appropriate network design or operation standards;
- (d) a developing database of precedents with similar events;
- (e) historical agreements between the participants;
- (f) the total cost impact.

Where the outcome of any dispute resolution proceedings would require or imply an amendment to the Code, the ERB shall refer the matter of amendment to the GCTC.

7 GRID CODE COMPLIANCE AUDITS

Any participant may request the GCTC to conduct an audit of another participant relating to compliance with part of, or the entire Code. The requesting participant may not request such information in relation to a particular section of the Code within six months of a previous request made under this section in relation to the relevant section. Where the participant to be audited believes that the request is frivolous or vexatious, they may declare a dispute under section 6.2.

A request under this section shall include the following information.

- (a) The nature of the request;
- (b) The name of the representative appointed by the requesting participant to conduct the investigation; and
- (c) The date and time or times at which the information is required.

A participant may not unreasonably withhold any relevant information requested. It shall be provided to the GCTC with such access to all relevant documentation, data and records (including computer records or systems) as is reasonably requested. Where the GCTC requests information that the participant considers confidential they shall maintain the confidentiality of that information.

The cost of such audits will be borne by the complainant if the audit reveals compliance and by the participant being audited if the audit reveals non-compliance.

CHAPTER 2 – NETWORK

1 APPLICATIONS FOR TRANSMISSION CONNECTIONS

All customers seeking connection or modification to the existing TS shall apply in writing to the TNSP.

The TNSP shall provide quotes for new connections or for upgrading existing connections, according to the ERB approved tariff methodology and within the following time frames:

Table 3: Connection time frames

	<i>Budget Quote</i>	<i>Firm Quote</i>
Connection service (business days)	≤30	≤60
Use of network service (days)	≤15	≤30

The customers may request provisional quote information from the TNSP that shall be provided without commitment and without detailed studies.

The agreed time period for connecting end-use-customers or upgrading connections shall be negotiated between the TNSP and the end-use-customers in every instance. The TNSP shall use the standard application form as shown in Appendix 6 for the processing of applications which should be read in conjunction with information provision requirements as specified in the Information Exchange Chapter.

End-use-customers shall enter into a connection contract and a PSA with the TNSP in advance of construction of the connection facilities.

End-use-customers shall enter into a use-of-system contract with the TNSP before the commencement of energy transactions over the TS.

2 CONNECTION CONDITIONS

This section specifies the minimum technical and design requirements that parties connected to or seeking connection to the TS shall adhere to.

2.1 Generator connection conditions

This section defines minimum requirements for generators connected to the TS, which are required to comply with the code.

Generators with capacity ratings as specified in Tables 4.1 (a) and 4.1 (b) below shall be subject to GCRs.

The SO shall evaluate and specify the need for optional IPS requirements which may not have been indicated in Tables 4.1 (a) and (b). The SO shall, on request, make available the requirements pertaining to the decision.

Table 4.1(a): Summary of the requirements applicable to specific ratings of non-Hydro Units.

Grid Code Requirement		Units other than Hydro (MW rating)		
		<20	20 to 100	101+
GCR1	Grid Protection			
	Backup Impedance	No	No	Yes
	Loss of Field	Yes	Yes	Yes
	Generator backup earth fault	Yes	Yes	Yes
	HV Breaker Fail	Yes	Yes	Yes
	HV Breaker Pole Discrepancy	Yes	Yes	Depends on System Requirements
	Unit Switch-on-to-standstill Grid Protection (achieved with controls)	Yes	Yes	
	Main Grid Protection only			Depends on System Requirements
	Main Grid Protection (with self-monitoring system) or main and backup			Depends on System Requirements
	Main and Backup Protection (with self-monitoring or monitoring systems)	Yes	Yes	
	Reverse Power	Yes	Yes	Yes
GCR2	Excitation system requirements	Yes	Yes	Yes
	Power System Stabilizer (To comment later)	—	—	
	Limiters (pressure, temp, speed available)	Yes	Yes	Yes
GCR3	Reactive Capabilities		Yes	Yes
GCR4	Governing	Yes	Yes	Yes
GCR5	Black Starting	Yes	No	No
GRC6	Emergency unit capabilities	Yes	Yes	Yes
GCR7	Independent action for control in system island	Yes	Yes	Yes

Table 4.1(b): Summary of the requirements applicable to specific classes of Hydro units

Grid Code Requirement		Hydro Units (MW rating)		
		<20	20 to 100	101+
GCR1	Protection			
	Backup Impedance	No	No	Yes
	Backup Impedance	Yes	Yes	Yes
	Loss of Excitation	Depends on System Requirements	Yes	Yes
	Gen transformer HV (backup) earth fault	Yes	Yes	Yes
	Gen TxHV REF protection	Yes	Yes	Yes
	HV Breaker Fail	—	Yes	Yes
	Breaker Pole Discrepancy	Yes	Yes	Yes
	Unit Switch-onto-standstill Protection		Depends on System Configuration	Depends on System Configuration
	Main Protection	Yes	Yes	The exact requirement will depend on System Configuration
	Reverse Power	Depends on System Configuration	Yes	Yes
GCR2	Excitation system requirements	Yes	Yes	Yes
GCR3	Reactive Capabilities	Depends on System Configuration	Yes	Yes
GCR4	Governing	Depends on System Requirements	Yes	Yes
GCR5	Black Starting	—	Yes	Yes
GCR6	Emergency unit capabilities	Depends on System Requirements	Depends on System Requirements	Yes
GCR7	Independent action for control in system island	—	—	Depends on System Requirements

The TNSP shall offer to connect and, subject to the signing of the necessary agreements, make available a point of connection to any requesting generator licensed to generate electricity.

2.1.1 Protection (GCR 1)

A generating unit, unit step-up transformer, unit auxiliary transformer, associated busbar ducts and switchgear shall be equipped with protection functions, maintained in line with international best practices, to rapidly disconnect appropriate plant sections should a fault occur within the relevant protection zones which fault may reflect into the TS.

Protection supplies for Main protection and other protection types shall be separately fused.

The IPS shall be protected with the following functions—

(a) Main protection

This protection shall be installed in the HV yard substation or in the unit protection panels. If this protection is installed in the unit protection panels then the DC supply for this protection and that used for control circuits shall be at least separately fused.

(b) Backup Impedance

An impedance facility with a large reach shall be used. This shall operate for phase faults in the unit, in the HV yard or in the adjacent transmission lines, with a suitable delay, for cases when the corresponding main protection fails to operate. The impedance facility shall have fuse fail interlocking.

(c) Loss of Field

All generating units shall be fitted with loss of field facility that matches the system requirements. The generator shall agree with the TNSP on the type of facility to be implemented.

(d) Generator Transformer HV back-up E/F Protection

This shall be an IDMT facility that shall monitor the current in the unit transformer neutral.

(e) HV Breaker Fail Protection

The “breaker fail” protection shall monitor the HV circuit breaker’s operation for protection trip signals, fault conditions.

(f) HV Pole discrepancy protection

The pole discrepancy protection shall cover the cases where one or two poles of a circuit breaker fail to operate after a trip or close signal.

(g) Unit Switch onto Standstill protection

This protection shall be installed in the HV yard substation or in the unit protection panels. If this protection is installed in the unit protection panels then the DC supply for this protection and that used for the circuit-breaker closing circuit shall be the same.

(h) Reverse Power

This protection shall be installed in the HV yard substation or in the unit protection panels. If this protection is installed in the unit protection panels then the DC supply for this protection and that used for the circuit-breaker closing circuit shall be the same.

General Considerations

In addition, should system conditions dictate, other protection requirements shall be determined by the SO in consultation with the generator and these should be provided and maintained by the relevant generator at its own cost.

Required HV breaker tripping, fault clearance times, including breaker operating times depend on system conditions and shall be defined by the TS.

All protection interfaces with the TNSP shall be co-ordinated between the participants.

The settings of all the protection tripping functions on the unit protection system of a unit, relevant to IPS performance and as agreed with each generator in writing, shall be co-ordinated with the transmission protection settings. These settings shall be agreed between the TNSP and each generator, and shall be documented and maintained by the generator, with the reference copy, which reflects the actual plant status at all time, held by the TNSP.

Protection setting documents shall illustrate plant capabilities and the relevant protection operations.

Prototype and routine testing shall be carried out as defined Appendix 7, section A.5.1

Any work on the protection circuits interfacing with transmission protection systems (e.g. bus zone) shall be sanctioned by the SO.

2.1.2 Excitation system requirements (GCR2)

A continuously acting automatic excitation control system shall be installed to provide constant terminal voltage control of the unit, without instability, over the entire operating range of the unit. (Note that this does not include the possible influence of a power system stabilizer.) Excitation systems shall comply with the requirements specified in IEC 60034.

The excitation control system shall be equipped with an under-excitation limiter, load angle limiter and flux limiter as described in IEC60034-16-1.

The excitation system shall have a minimum excitation ceiling limit of 1.6 pu rotor current, where 1 pu is the rotor current required to operate the unit at rated load and at rated power factor.

The settings of the excitation system shall be agreed between the SO and each generator, and shall be documented, with the master copy held by the SO. The generators shall control all other copies. The procedure for this is shown in Appendix 7, Section A.5.2.

In addition, the unit shall be capable of operating in the full range as indicated in the capability chart supplied as part of Appendix 15. Test procedures are shown in Appendix 7, Section A.5.5.

Power system stabilizers as described in IEC60034-16-1 are a requirement for all new units, and for existing units retrofitting may be required depending on IPS requirements. The requirements for other excitation control facilities and AVR refurbishment shall be determined in conjunction with the SO.

Routine and prototype response tests shall be carried out on excitation systems as indicated in Appendix 7, Section A.5.2 and in accordance with IEC60034-16-3.

2.1.3 Reactive capabilities (GCR3)

All new units shall be capable of supplying rated power output MW at any point between the limits 0.85 power factor lagging and 0.95 power factor leading at the unit terminals.

Reactive output shall be fully variable between these limits under AVR, manual or other control.

Routine and prototype response tests shall be carried out to demonstrate reactive capabilities as indicated in Appendix 7, Section A.5.3.

2.1.4 Governing (GCR4)

2.1.4.1 Design Requirements

The SAPP Operating Sub-committee approved relaxed frequency bands in the power pool from $\pm 0.05\text{Hz}$ to $\pm 0.15\text{Hz}$ during normal operations¹. All units above 50 MVA shall have an operational governor that shall be capable of responding according to the minimum requirements set out in this document.

2.1.4.2 System Frequency Variations

The nominal frequency of the TS is 50Hz and is normally controlled within the limits of $\pm 0.15\text{Hz}$ unless exceptional circumstances prevail. The system frequency could rise or fall in exceptional circumstances and generating units must be capable of continuous normal operation for the high and low frequency conditions set out in section 2.1.5 when the TS comes out of synchronism with the SAPP network.

Design of turbo-alternator units must enable continuous operation, at up to 100% active power output, within this range.

Hydro-alternator units must be capable of continuous normal operation for high frequency conditions described in section 2.1.5.4 and low frequency conditions as described in 2.1.5.6.

2.1.4.3 HighFrequency Requirements for Turbo-alternators

All synchronized units shall respond by reducing active power to frequencies above 50 Hz plus allowable limits described in section 2.1.5.2. Speed governors shall be set to a predetermined droop characteristic. The response shall be fully achieved within the time set in the dynamic study of the system and must be sustained for the duration of the frequency excursion. The unit shall respond to the full designed minimum operational capability of the unit at the time of the occurrence.

Over-frequency Conditions above 50.15Hz

The frequency requirements and behaviour of turbo generators will be set out in the dynamics study, which will be commissioned by the GCTC.

Turbo generators will be equipped with over frequency protection at a set value, but this value shall not be set at a level likely to compromise the system security and safety.

2.1.4.4 High Frequency Requirements for Hydro Alternators

All synchronised hydro units shall respond by reducing active power to frequencies above 50 Hz plus allowable limits described in section 2.1.4.7. Speed governors shall be set to give a predetermined droop characteristic. The response shall be fully achieved within the stipulated time and must be sustained for the duration of the frequency excursion. The unit shall respond to the full load capability range of the unit.

Hydro generators will be equipped with over frequency protection at a set value, but this value shall not be set at a level likely to compromise the system security and safety.

2.1.4.5 Low Frequency Requirements for Turbo-alternator Units

Low frequency response is defined as an ancillary service in the form of operating reserve. However all units shall be designed to be capable of having a droop characteristic with a preset minimum response within a certain time of a frequency incident within which it must be sustained.

Under frequency protection shall be equipped at preset levels, which shall be coordinated with automatic load shedding.

Low frequency below 49.85Hz

The behaviour of generating units will be determined by a dynamics study to be commissioned by the GCTC.

2.1.4.6 Low Frequency Requirements for Hydro-alternator Units

All reasonable efforts shall be made by the generator to avoid tripping of the hydro-alternator for under frequency conditions provided that the system frequency is above 47.5Hz. At this frequency, the IPS has been set to separate with all the interconnectors at a preset time delay. This value is meant to preserve system integrity.

If the system frequency falls below 47.5Hz for more than 1 second, independent action may be taken by a generator to protect the unit. Such action includes ability to islanding and independent action in island position.

2.1.4.7 Dead band

The maximum allowable dead band shall be ± 0.15 Hz for governing. That is, no response is required from the unit while the frequency is greater than 49.85 and less than 50.15 Hz.

Routine and prototype response tests shall be carried out on the governing systems as indicated in Appendix 7, Section A.5.4

2.1.5 Black starting (GCR5)

Power stations that have declared that they have a station black start capability shall demonstrate this facility by test as described in Appendix 7, Section A.5.5.

Black start capable power stations may be called from time to time to carry out a black start as described in Appendix 7, Section A.5.5.

2.1.6 Emergency unit capabilities (GCR6)

All generators shall specify their units' capabilities for providing emergency support under abnormal power system conditions, as detailed in the System Operation Chapter.

2.1.7 Facility for independent generator action. (GCR7)

Frequency control under system island conditions shall revert to the power stations as the last resort, and units and associated plant shall be equipped to handle such situations. The required control range is from 49.85Hz to 50.15Hz.

2.1.8 Testing and compliance monitoring

A generator shall keep records relating to the compliance by each of its units with each section of this code applicable to that unit, setting out such information that the SO reasonably requires for assessing power system performance (including actual unit performance during abnormal conditions).

A generator shall review, and confirm to the SO, compliance by each of that generator's units with every GCR as specified in Appendix 6.

A generator shall conduct tests or studies to demonstrate that each power station and each generating unit complies with each of the requirements of this code. Tests shall be carried out on new units, after every outage where the integrity of any GCR may have been compromised, to demonstrate the compliance of the unit with the relevant GCR or GCRs. The generator shall continuously monitor its compliance in all material respects with all the connection conditions of the Code.

Each generator shall submit to the SO a detailed test procedure, emphasizing system impact, for each relevant part of this code prior to every test.

If a generator determines, from tests or otherwise, that one of its units or power stations is not complying with any material respect of one or more sections of this code, then the generator shall —

- (a) promptly notify the SO of that fact;
- (b) promptly advise the SO of the remedial steps it proposes to take to ensure that the relevant unit or power station, as the case may be, can comply with this code and the proposed timetable for implementing those steps;
- (c) diligently take such remedial action as will ensure that the relevant unit or power station, as the case may be, can comply with this code. The generator shall regularly report in writing to the SO on its progress in implementing the remedial action;
- (d) after taking remedial action as described above, demonstrate to the reasonable satisfaction of the SO that the relevant unit or power station, as the case may be, is then complying with this code.

2.1.9 Non-compliance suspected by the SO

If at any time the SO believes that a unit or power station is not complying with this code, the SO may —

- (a) notify the relevant generator of such non-compliance specifying the code section concerned and the basis for the SO's belief; or
- (b) issue an instruction requiring a generator to carry out a test to demonstrate that the relevant power station complies with the GCRs. A generator shall not refuse such an instruction unless it is not issued timeously and there are no reasonable grounds for suspecting non-compliance.

However, if the relevant generator believes that the unit or power station, as the case may be, is complying with the code, then the SO and the generator shall promptly meet to resolve their difference.

2.1.10 Unit modifications

(a) Modification proposals

If a generator proposes to change or modify any of its units in a manner that could reasonably be expected to either adversely affect that unit's ability to comply with this code, or changes the performance of, information supplied to, or settings of, that unit, the generator shall submit a proposal notice to the SO which shall:

- (i) contain detailed plans of the proposed change or modification;
- (ii) state when the generator intends to make the proposed change or modification; and
- (iii) set out the proposed tests to confirm that the relevant unit as changed or modified operates in the manner contemplated in the proposal and in compliance with this code.

If the SO and the generator disagree on the proposal submitted, the two shall refer the matter to the ERB for resolution.

(b) Implementing and Testing modifications

A generator shall ensure that an approved change or modification to a unit or to a subsystem of a unit is implemented in accordance with the relevant proposal approved by the SO.

A generator shall notify the SO promptly after an approved change or modification to a unit has been implemented, and confirm that it conforms to the relevant proposal by conducting the relevant tests, in relation to the connection conditions.

2.1.11 Equipment requirements

Where a generator needs to install equipment that connects directly with TNSP equipment, such as in the HV yard of the TNSP, such equipment shall adhere to the TNSP design requirements as set out in this code.

The TNSP may require end-use customers to provide documentary proof that their connection equipment complies with all relevant standards, both by design and by testing.

2.2 Distributors and end-use customers

This section describes connection conditions for distributors and end-use customers.

The TNSP shall offer to connect and, subsequent to the signing of the relevant agreements as stated in Section 1, make available a point of connection to any requesting distributor or end-use customer.

2.2.1 Power factor

Distributors and end-use customers shall take all reasonable steps to ensure that the power factor at the point of supply is at all times 0.92 lagging or better, unless otherwise agreed to in existing contracts between the participants. This requirement also applies to each point of supply individually for distributors and end-use customers with more than one point of supply. A leading power factor shall not be acceptable, unless specifically agreed to in writing.

A penalty and incentive scheme shall be agreed between the TNSP and the distributor or end-use customer regarding the limits of the power factor. Should the power factor be out of the agreed limits, the penalty and incentive scheme shall apply. The participants shall co-operate in determining the plans of action to rectify the situation on a permanent basis. Overall lowest cost solutions shall be sought.

2.2.2 Protection

A participant shall take reasonable steps to protect the participant's plant in accordance with prudent utility practice.

The SO protection requirements, with which the distributors and end-use customers shall interface, are described in Section 4. The relevant participants, insofar as the equipment of one participant may have an impact on the other, shall agree to the detailed protection applications, in writing. Distributors and end-use customers connected directly to the transmission substations shall ensure that they comply with the protection standards of the relevant TNSP.

The participants shall co-ordinate protection to ensure proper grading and protection coordination.

Protection dependability (IEC50-448) shall not be less than 99%.

2.2.3 Fault levels

Distributors and end-use customers shall ensure that their equipment is capable of operating at specified fault levels as published by the SO from time to time.

In the event that works are carried out that will result into fault level ratings exceeding that of distributor or end-use customer equipment, the TNSP shall seek overall lowest cost solutions to address the problem. Corrective action shall be for the cost of the relevant asset owner that has caused the increase in fault levels.

2.2.4 Network Performance

If the distributor or end-use customer network performance falls below the specifications of ZS387 Electricity Supply – Power Quality and Reliability standard, or affects the QOS to other distributors and end-use customers, or causes damage, direct or indirect to the TNSP equipment, the process for dispute resolution, as described in the Governance Chapter, shall be followed.

If distributors or end-use customers are aware that their network performance could be unacceptable as described above, they shall take reasonable steps at their own cost to overcome the shortcomings. These changes should be effected in consultation with the TNSP on both the technical scope and the time frame.

2.2.5 Equipment requirements

Where the distributor or end-use customer needs to install equipment that connects directly with TNSP equipment in transmission substations, such equipment shall adhere to the TNSP design requirements as set out in Section 3.

The TNSP may require end-use customers to provide documentary proof that their connection equipment complies with all relevant design requirements, both by design and by testing.

2.2.6 Additional Re-inforcement

An end-use customer may request additional reinforcements to the TS over and above that which could be economically justified as described in the section on TS Planning and Development. The relevant TNSP shall provide such reinforcements if the end-use customer agrees to bear the costs, which shall be priced according to the negotiated contract.

3 TNSP TECHNICAL DESIGN REQUIREMENTS

The purpose of this section is to document the design and other technical standards that the TNSP shall adhere to.

3.1 *Equipment design standards*

Primary substation equipment shall comply with the relevant IEC specifications. The TNSP shall ensure that the agreed design standards at the end-use customer interface are documented. This documentation shall address the interface for the primary equipment and secondary circuits. Consideration shall also be given to possible common DC supplies, AC supplies, compressed air systems and fencing.

The TNSP shall ensure that switching devices at or near a power station are adequately rated and capable of switching loads and fault currents without generating undue switching surges. Switching surge protection provided to the generating unit should be as provided by IEC60099.

End-use customers may require the TNSP to provide documentary proof that their connection equipment complies with all relevant standards, both by design and by testing.

3.2 *Clearances*

Safety clearances from TS equipment shall at least comply with ZS415 Electrical Safety Code – Code of Practice.

3.3 *CT and VT ratios and cores*

The asset owner in consultation with the other relevant participants shall determine CT and VT ratios and cores.

A TNSP, distributor or an end-use customer connected to the TS shall ensure that measurement equipment complies with the accuracy classes specified in Electricity metering – Code of Practice: ZS 647 for the purpose of operation and control of the IPS.

3.4 *Standard busbar arrangements and security criteria*

The standard substation arrangement shall be based on providing one busbar zone for every main transformer or line normally supplying that busbar. System reliability criteria as described in the TS Planning and Development Section should also be adhered to.

3.5 *Motorised Isolators*

In the interest of safety, all new substations with voltage levels of above 132kV shall be provided with motorized isolators. Substations with voltages equal to or lower than 132kV may be fitted with motorized isolators depending on the merits of cost and operations.

3.6 *Earthing and Surge Protection*

The TNSP shall ensure adequacy of all earthing installations to provide for—

- (a) safety of personnel and the environment;
- (b) correct operation of all protection systems; and
- (c) agreed design and performance levels.

3.6.1 Earthing isolators

Earthing isolators shall be provided at new substations where the fault level is designed for 20 kA and above.

The TNSP shall provide adequate protection to limit lightening surges at the connection point as specified in IEC 60071 and IEC 60099.

3.7 Tele-control

A participant may be permitted to have tele-control equipment in the substations yards, or buildings of the other party, to perform agreed monitoring of own equipment and other mutually agreed equipment. Access shall be provided to such equipment.

3.8 Substation drawings

The following minimum set of drawings shall be made available by the respective asset owners for all points of supply, if required by the other party for the purposes of connection—

- (a) Key Plan;
- (b) Bay Layout Schedules;
- (c) Foundation, Earth-mat/ Earth-pit and Trench Layout;
- (d) Steelwork Marking Plan;
- (e) Security Fence Layout;
- (f) Terrace, Road and Drainage Layout;
- (g) Transformer Plinth;
- (h) General Arrangement;
- (i) Drawings with Sectional details;
- (j) Slack Span Schedule;
- (k) Barrier Fence Layout;
- (l) Security Lighting;
- (m) Floodlighting Parameter Sketch;
- (n) Protection details;
- (o) Contour Plan; and
- (p) Station Electric Diagram.

All drawings shall use the standard electrical symbol set defined in Appendix 8.

3.9 Recorders

The TNSP shall install QOS recorders as stipulated by ZS387 Electricity Supply-Power Quality and Reliability standard at points of supply or points of connection.

The TNSP shall install disturbance recorders at locations in the network that shall enable the SO to adequately analyze system disturbances.

Access to the records shall be as specified in the Information Exchange Chapter.

3.10 *Fault Levels*

The TNSP shall maintain contracted minimum fault levels at each point of supply.

Where mitigating actions have been put in place to reduce fault levels, the SO shall document maximum fault levels, before and after such actions.

The TNSP and SO shall liaise with distributors and end-use customers on how fault levels are planned to change and on the best overall solution when equipment ratings become inadequate. The TNSP shall communicate the potential impact on the safety of people when equipment ratings are exceeded.

3.11 *The TNSP's delivered QOS*

The TNSP shall comply with ZS387 Electricity Supply – Power Quality and Reliability standard at every point of supply.

Where the TNSP fails to meet the standard parameters, it shall take reasonable steps at own cost to overcome the shortcomings. These changes should be effected in consultation with the distributor or end-use customer on both the technical scope and the time frame.

4 SERVICE PROVIDER PROTECTION REQUIREMENTS

This section specifies the minimum protection requirements for TNSPs as well as typical settings, to ensure adequate performance of the TS as experienced by the distributors and end-use customers.

The TNSP shall conduct periodic or CBM testing of equipment and systems to ensure and demonstrate that these are performing to the design specifications.

The TNSP shall make available to end-use customers results of tests performed on equipment.

4.1 *Equipment protection requirements*

4.1.1 Feeder protection: above 132kV

4.1.1.1 Protection Design Standards

New feeders shall be protected by two main protection systems – Main 1 and Main 2 of which one shall be line differential and the other distance protection. The Main 1 and Main 2 protection systems shall be fully segregated in secondary circuits.

An additional DEF shall be incorporated in the main protection relays or installed separately to alleviate possible deficiencies of distance relays in detection of high resistance faults.

Non-directional O/C and E/F protection functions shall be installed either separately or as integral parts of the main protection relay to provide optional backup. This could use definite time and IDMT over-current.

4.1.1.2 Protection Settings

The protection relays shall provide reliable protection against all possible short circuits and remote or local back up for busbar faults that are not cleared and those that are not set to provide overload tripping. Where specifically required, the feeder protection may be set to provide remote back up for other faults as agreed upon with other participants.

4.1.1.3 Automatic Re-closing

ARC facilities shall be provided on all feeders.

The SO shall decide on ARC selection based on real time system, environmental constraints and consultation with distributors and end-use customers, with regard to equipment capabilities and in accordance with the ARC philosophy in this section. All ARC settings and methodology shall be implemented by the TNSP and be made available to distributors and end-use customers on request.

4.1.1.4 Power Swing Blocking

New distance relays on the TS shall be equipped with PSB facility. Unwanted operations of distance relays during power swing conditions shall be blocked on the TS. On Control Area tie lines, the philosophy implemented under SAPP shall prevail.

4.1.2 Feeder protection: 66kV and below, at TNSP substations

4.1.2.1 Design Standard

These feeders shall be protected by a single protection system, incorporating either distance or differential protection relays, unless otherwise agreed. Definite time and Inverse Definite Minimum Time (IDMT) over-current shall provide back up.

The protection shall be equipped with automatic re-closing. Synchronising relays shall be provided on feeders that operate in "ring supplies" and are equipped with line voltage transformers.

4.1.2.2 Protection Settings

Protection relays shall provide reliable protection against all possible short circuits as well as remote or local rack up for uncleared busbar faults and should not be set to provide time delayed overload tripping where measurements and alarms are provided on the SCADA system.

4.1.2.3 Automatic Re-closing

The distributor or end-use customer shall determine ARC requirements. The SO may specify additional ARC requirements for system security reasons, which could extend beyond the TNSP substations.

4.1.3 Tele-protection requirement:

New distance protection systems shall be equipped with tele-protection facilities to enhance the speed of operation.

4.1.4 Transformer and reactor protection

The standard schemes for transformer protection shall be designed to provide the requisite degree of protection for the following fault conditions:

- (a) faults within the tank;
- (b) faults on transformer connections;
- (c) overheating; and
- (d) faults external to the transformer.

The TNSP shall consider the application of the following schemes in the design of the protection system:

- (a) Transformer IDMT E/F;
- (b) Transformer HV or MV IDMT O/C;
- (c) Transformer HV/MV Instantaneous O/C;
- (d) Transformer LV (Tertiary) IDMT or Instantaneous O/C;

- (e) Transformer Current Differential Protection;
- (f) Transformer High Impedance Restricted E/F;
- (g) Transformer Thermal Overload; and
- (h) Buchholtz alarm and trip protection.

4.1.5 Transmission Busbar Protection

Busbars shall be protected by bus-zone current differential protection set to be as sensitive as possible for the "in-zone faults" and maintain stability for any faults outside the protected zone, even with fully saturated CT.

4.1.6 Transmission bus coupler and bus section protection

Bus-coupler and bus-section panels are equipped with O/C protection.

4.1.7 Transmission shunt capacitor protection

All the new high voltage capacitor banks shall be equipped with sequence switching relays to limit inrush current during capacitor bank energisation. Inrush reactors and damping resistors shall also be employed to limit inrush current.

The following protection functions shall be provided for all types of protection schemes:

- (a) unbalanced protection with alarm and trip stages;
- (b) O/C protection with instantaneous and definite time elements;
- (c) E/F protection with instantaneous and definite time sensitive function;
- (d) overload protection with IDMT characteristic;
- (e) over voltage with definite time;
- (f) circuit breaker close inhibit for 300 seconds after de-energisation; and
- (g) ancillary functions as indicated below.

4.1.8 Over-voltage protection

Primary protection against high transient over-voltages of magnitudes above 140%, such as those induced by lightning, shall be provided by means of surge arrestors. To curtail dangerous, fast developing over-voltage conditions that may arise as a result of disturbance, additional over-voltage protection shall be installed on shunt capacitors and feeders.

Over-voltage protection on shunt capacitors is set to disconnect capacitor at 110% voltage level with a typical delay of 200 milliseconds to avoid unnecessary operations during switching transients.

Over-voltage protection on the feeders is set to trip the local breaker at voltage level of 120% with a delay of 1 to 2 seconds.

4.1.9 Ancillary Protection Functions

Protection systems are equipped with auxiliary functions and relays that enable adequate co-ordination between protection devices and with bay equipment. The TNSP shall consider the following functions for all new protection system designs:

- (a) Breaker Fail and Bus trip;
- (b) Breaker Pole Discrepancy;
- (c) Breaker Anti-pumping;
- (d) Pantograph Isolator Discrepancy (Pole discrepancy applied to Isolators); and
- (e) Master trip and lockout.

4.2 *System protection requirements*

4.2.1 Under-frequency load shedding

The actions taken on the power system during an under-frequency condition is defined in the System Operation Chapter.

Under-frequency load shedding relays shall be installed in the IPS as determined by the SO in consultation with distributors and end-use customers. The respective asset owners shall pay for the installation and maintenance of these relays.

Under-frequency relays shall be tested periodically. Distributors and end-use customers shall submit to the SO a written report of each such test, within a month of the test being done, in the format specified in the Information Exchange Chapter.

The provisions of the SAPP operating guidelines shall prevail over the provisions of this section in the event of conflict.

4.2.2 Out-of-step tripping (OST)

The purpose of the OST protection is to separate power system in a situation where a loss of synchronous operation takes place between a unit or units and the main power system. In such a situation system separation is desirable to remedy the situation. Once the islanded system is stabilized it can be reconnected to the main system.

The SO shall determine and specify the OST functionality to be installed at selected locations by the TNSP.

4.2.3 Under-voltage load shedding

Under-voltage load shedding protection schemes are used to prevent loss of steady-state stability under conditions of large local shortages of reactive power (voltage collapse). Automatic load shedding tripping of suitable loads is carried out to arrest the slide.

The SO shall determine and specify the under voltage load shedding functionality to be installed at selected locations by the TNSP.

4.3 *Protection system performance monitoring*

The TNSP shall monitor protection performance.

Each protection operation shall be investigated for its correctness based on available information. The TNSP shall provide a report to distributors and end-use customers affected by a protection operation when requested to do so.

5 NOMENCLATURE

All safety terminology shall comply with the ZS418 Electrical Safety – Code of Practice. Engineering drawings relating to connecting equipment shall use the symbol set and layout conventions as defined in Appendix 8.

6 TS PLANNING AND DEVELOPMENT

This section specifies the technical, design and economic criteria and procedures to be applied by the TNSP in the planning and development of the TS and to be taken into account by distributors and end-use customers in the planning and development of their own systems. It specifies information to be supplied by distributors and end-use customers to the TNSP, and information to be supplied by the TNSP to distributors and end-use customers.

6.1 *Planning process*

The TNSP shall follow a planning process divided into major activities as follows—

- (a) identification of network requirements;
- (b) formulation of alternative options to meet this need;
- (c) studying these options to ensure compliance with agreed technical limits, and justifiable reliability and QOS standards;
- (d) costing these options on the basis of present-day capital costs and using appropriate net discount rates, establish the net present cost of each option;
- (e) determining the preferred option;
- (f) building a business case for the preferred option using the approved justification criteria; and
- (g) requesting approval of preferred option and initiating execution.

6.2 *Forecasting the demand*

The SO is responsible for producing the TS demand forecast for the next fifteen years and updating it annually.

The TS demand forecast shall be determined for each point of supply.

The SO shall forecast the maximum demand in MW for each transmission substation and the result shall be reconciled with data from participants.

Distributors and end-use customers shall annually, by end of October, supply their 15-year-ahead load forecast data as detailed in the Information Exchange Chapter.

6.3 *Technical limits and targets for planning purposes*

The limits and targets against which proposed options are checked by the TNSPs shall include technical and statutory limits, which must be observed.

6.3.1 *Voltage limits and targets*

Technical limits are stated in Appendix 9.

6.3.2 *Other targets for planning purposes*

6.3.2.1 *Transmission Lines*

Thermal ratings of standard transmission lines shall be determined and updated from time to time. The thermal ratings shall be used as an initial check of line overloading. If the limits are exceeded the situation shall be investigated as it may be possible to defer strengthening depending on the actual line and on local conditions.

6.3.2.2 Transformers

Standard transformer ratings shall be determined and updated from time to time using IEC specifications. If target loads are exceeded the specific situation shall be assessed as it may be possible to defer adding extra transformers.

6.3.2.3 Series Capacitors

With the system healthy, the maximum steady state current should not exceed the rated current of the series capacitor.

Cyclic overload capabilities shall follow the provisions of IEC 143

The particular rating to be used must match the duration of the contingency with the required overload capability. Duration of contingency will depend on ability to pick up generation or shed load and the load profile.

A TNSP wishing to install a new series capacitor or modify the size of an existing series capacitor, shall arrange for sub synchronous resonance, harmonic and protection coordination studies to be conducted to ensure that sub synchronous resonance will not be excited in any generator.

6.3.2.4 Shunt Reactive Compensation

Shunt capacitors shall be able to operate at 30% above their nominal rated current at nominal voltage to allow for harmonics and also for voltages up to the maximum level allowable.

A participant wishing to install or modify such equipment shall at the participant expense arrange for harmonic resonance studies to be conducted. If such studies indicate possible harmonic resonance conditions, which could impact on the TS, such party shall inform the SO before proceeding with the installation or modification.

6.3.2.5 Circuit Breakers

These ratings, and the following limits specified for circuit breakers, shall not be exceeded:

- (a) single-phase breaking current: 1.15 times 3 phase fault current; or
- (b) peak making current: 2.55 times 3 phase rms fault current.

6.3.2.6 Secondary ARC current during single-phase reclosing

The secondary arc current shall not exceed —

- (a) 20 amps rms with recovery voltage of 0.4 pu; or
- (b) 40 amps rms with recovery voltage of 0.25 pu.

6.3.3 Reliability criteria for planning purposes

The TNSPs shall formulate long-term plans for expanding or strengthening the TS on the basis of the justifiable redundancy.

6.3.4 Contingency criteria for planning purposes

A system meeting the n-1 or n-2 contingency criterion must comply with all relevant limits outlined in Appendix 9 and the applicable current limits, under all credible system conditions.

For contingencies under various loading conditions it shall be assumed that appropriate, normally used, generating plant is in service to meet the load and provide spinning reserve. For the more probable n-1 network contingency the most unfavourable generation pattern within these limitations shall be assumed, while for the less probable n-2 network contingency an average pattern shall be used. More details of load and generation assumptions for load flow studies are given in section 6.3.5.6.3.5

6.3.5 Integration of Power Stations

When planning the integration of power stations the following criteria shall apply:

6.3.5.1 Power stations of less than 1000 MW—

- (a) With all connecting lines healthy, it shall be possible to transmit the total output of the power station to the system for any system load condition. If the local area depends on the power station for voltage support, connection shall be done with a minimum of two lines;
- (b) Transient stability shall be maintained following a successfully cleared single phase fault;
- (c) If only a single line is used it shall be able to be selected to alternative busbars and be able to go on to bypass at each end of the line.

6.3.5.2 Transient stability

Transient stability shall be retained for the following conditions:

- (a) a three-phase, line or busbar fault, cleared in normal protection times, with the system healthy and the most onerous power station loading condition;
- (b) a single phase fault cleared in "bus strip" times, with the system healthy and the most onerous power station loading condition; or
- (c) a single-phase fault, cleared in normal protection times, with any one line out of service and the power station loaded to average availability.

6.3.5.3 Busbar arrangements

Bus-bar layouts shall allow for selection to alternative bus-bars and the ability to go on to by-pass.

6.3.6 Least economic cost criteria

When investments are made in terms of improved supply reliability or quality, this would be the preferred method to use. This methodology should also be used to determine or verify the desired level of network or equipment redundancy. The methodology requires that the cost of poor network services needs to be determined. These include the cost of interruptions, load shedding, network constraints, poor quality of supply, etc. Statistical analysis of network outages is also required.

The least-cost investment criterion equation to be satisfied can be expressed as follows:

$\text{Value of improved QOS to } > \text{Cost to the service provider to end use customers provide improved QOS}$
--

From the equation above it is evident that if the value of the improved QOS to the distributor or end-use customer is less than the cost to the service provider, then the service provider should not invest in the proposed project(s).

Equation above can be stated differently as:

Annual value (US\$/kWh) x Reduction in > Annual cost to the service EENS to consumers (kWh) provider to reduce EENS (R)

The reduction in EENS is calculated on a probabilistic basis based on the improvements derived from the investments

The cost of unserved energy is a function of the type of load, the duration and frequency of the interruptions, the time of the day they occur, whether notice is given of the impending interruption, the indirect damage caused, the start-up costs incurred by the consumers, the availability of end-use customer back-up generation and many other factors.

Appendix 10 indicates the concept of a load profile, while Appendix 11 indicates the energy not served. This is in the event of a load growing to 125 MVA whereas the firm transformer rating is 100 MVA.

6.3.7 Cost reduction investments

Proposed expenditure which is intended to reduce service providers' costs (for example, shunt capacitor installations, telecommunication projects and equipment replacement which reduce costs, external telephone service expenses and maintenance costs respectively) or the cost of losses or other ancillary services, should be evaluated in accordance with this section.

It is necessary to calculate the NPV of the proposed investment using the DCF methods. This should be done by considering all cost reductions such as savings in system losses as positive cash flows, off-setting the required CAPEX. A sensitivity analysis with respect to the amount of CAPEX (estimated contingency amount), the AAICOG (when appropriate) and, future load growth scenarios is required. A resulting positive NPV indicates that the investment is justified over the expected life of the proposed new asset.

However, a positive NPV does not always indicate the optimal timing for the investment. For this reason, the second portion of the cost reduction analysis is necessary. It remains necessary to ascertain if the annual extra costs incurred by the service provider for owning and operating the proposed asset is less than all cost reductions resulting from the new asset in the first year that it is in service.

6.3.8 Statutory or strategic investments

This category of projects includes the following:

- (a) investments formally requested by the Government of the Republic of Zambia;
- (b) projects necessary to meet environmental legislation;
- (c) possible compulsory contractual commitments; and
- (d) servitude acquisition.

This category shall not be used for justifying projects that are merely not of economic benefit.

6.4 Development investigation reports

The TNSPs shall compile, before any development of the TS is approved, a detailed development investigation report. The report shall be used as the basis for making the investment decision and shall as a minimum contain the following elements—

- (a) a description of the problem or request;
- (b) alternatives considered, including non-transmission or capital and an evaluation of the long-term costs/benefits of each alternative; and
- (c) detailed techno-economic justification of the selected alternative according to the approved investment criteria.

6.5 *Transmission investment plan*

The TNSPs shall publish, annually a five-year-ahead network investment plan, indicating the major capital investments planned, but not yet necessarily approved.

This plan shall be based on all end use customer requests received at that time, as well as TNSP's initiated projects.

6.6 *Mitigation of network constraints*

The TNSPs have the obligation to resolve network constraints.

Network constraints or "congestion" shall be regularly reviewed by the TNSPs. Economically optimal plans shall be put in place around each constraint which plans may involve investment, the purchase of the constrained generation ancillary service or other solutions.

6.7 *Interfacing between participants and TNSPs*

The TNSPs shall ensure regular interfacing with distributors and end-use customers regarding network development. One objective shall be to achieve overall optimal plans, ensuring economically efficient investment.

The TNSPs shall coordinate their system planning with the objective of achieving overall optimal plans hence ensuring economically efficient investment.

6.8 *Special end use customer requirements for increased reliability*

Should an end-use customer require a more reliable connection than the one provided by the TNSPs, and if the end-use customer is willing to pay the total cost of providing the increased reliability, the TNSP shall meet the requirements at the lowest overall cost.

7 *NETWORK MAINTENANCE*

Participants shall monitor the performance of their plant and take appropriate action where deteriorating trends are detected.

Maintenance scheduling shall be done in accordance with the System Operation Chapter.

The TNSP shall agree, in writing, with its end-use customers, details of any special maintenance requirements as well as maintenance coordination requirements per transmission substation. The TNSP shall provide distributors and end-use customers with details of its maintenance plans and practices upon request.

CHAPTER 3 – METERING

1 OBJECTIVE

The Metering Chapter specifies tariff and statistical metering requirements for the ZAGC.

2 SCOPE

This chapter shall apply to metering points at the boundary of the TNSPs and their customers, TNSP -TNSP boundaries and international boundaries; and it shall be applicable to—

- (a) main metering installations and check metering installations used for the measurement of active and reactive energy;
- (b) the collection and storage requirements for metering data;
- (c) the provision, installation, maintenance and testing of metering equipment;
- (d) the standards and accuracy of all metering equipment used in the process of electricity metering; and
- (e) the relationship of entities involved in the electricity metering industry.

Where SAPP requirements and this Code have similar provisions in relation to metering installations, the stricter of the two shall apply.

Where there are requirements on the provision of meter data under the SAPP as well as under this Code both sets of data provision shall be obeyed.

The requirements of this metering chapter do not apply to operational metering on the IPS as utilised in the SCADA systems.

3 GENERAL PROVISIONS

The metering point shall be located at commercial boundary between the participants.

Customers may request an independent audit of metering installations. The requesting participant shall be responsible for any costs relating to the audit unless the metering installation is proved to be outside the defined standard.

Parties shall provide all the necessary information reasonably required to perform their respective metering duties.

The type, installation, operation and testing of metering installation at each metering point shall comply with ZS 647 metering specifications.

4 RESPONSIBILITY FOR METERING INSTALLATIONS

The Service Provider shall ensure that all points identified as metering points in accordance with Section 3 have metering installations in accordance with the provisions of this code.

The TNSP participants shall be responsible for managing and collecting metering information.

Participants connected to or intending to connect to the TS shall provide the relevant service provider with all information deemed necessary to enable performance of its metering duties.

The participants shall ensure that commissioning, maintenance, auditing and testing of metering installations are carried out in accordance with ZS 647.

5 METERING INSTALLATION REQUIREMENTS

- (a) each metering point shall be installed with main and check metering;
- (b) both active and reactive energy shall be measurable without compromising any requirements of this code;
- (c) full four quadrant metering shall be installed where active and reactive energy flow is in both directions;
- (d) the meters shall be configured to store or record metering data in half hourly integration periods. This must be in non-volatile memory which is able to be maintained without external power;
- (e) the meter or meters, or recorder or recorders shall be able to store data in memory for at least 40 days;
- (f) the service provider shall provide historical metering data to the customers on request. Note ZS 647 stipulates 5 years data retention;
- (g) the service provider shall provide real time metering data when requested to do so by the customer provided that this is practicably possible. The customer shall bear the cost of this installation; and
- (h) a customer may install own metering in a service provider's installation. Customer metering is secondary metering only. This provision is subject to there being no impact on the settlement metering.

6 DATA VALIDATION

The participants shall carry out data validation in accordance with ZS 647 in the event of—

- (a) electronic access to the meters not being possible;
- (b) an emergency bypass or other scheme having no metering system; or
- (c) metering data not being available.

The participant's officer responsible for metering may in carrying out data validation resort to any of the following—

- (a) manual meter data downloading;
- (b) estimation or substitution subject to mutual agreement between the affected parties;
- (c) profiling; or
- (d) reading of the meter at scheduled intervals.

When the need for metering estimations arises, the estimates will be done in accordance with ZS 647.

7 METER VERIFICATION

In addition to the ZS 647 verification requirements, meter readings shall be compared with the metering database at least once a year.

8 METERING DATABASE

The service provider shall create, maintain and administer a metering database containing in addition to meter measurement data the following information—

- (a) name and unique identifier of the metering installation;
- (b) the date on which the metering installation was commissioned;
- (c) the connecting parties at the metering installation;
- (d) maintenance history schedules for each metering installation;
- (e) telephone numbers used to retrieve information from the metering installation;
- (f) type and form of the meter at the metering installation;
- (g) fault history of a metering installation; and
- (h) commissioning documents for all metering installations.

The service provider shall retain metering information for at least five years for audit trail purposes.

9 METERING DATABASE INACCURACIES

In the event of test results revealing that data in the metering database is inaccurate, the service provider shall inform all affected participants and corrections shall be made to the official metering data and the associated billing by mutual agreement.

10 ACCESS TO METERING DATA

The service provider shall make metering data available to affected participants in the format agreed upon.

Service providers shall publish all data in the formats agreed upon by participants. New formats shall be negotiated between the service provider and the affected participants. The service provider shall store all customer information in a database and shall ensure that the database is maintained and updated continuously.

The service provider shall at the expense of the requesting party, provide non-standard data provision methods.

Security requirements for metering data shall be as specified in ZS 647.

11 CONFIDENTIALITY

Metering data for use in energy trading and billing are confidential information and shall be treated in accordance with the Information Exchange Chapter.

CHAPTER 4 – SYSTEM OPERATION

1 OBJECTIVE

The objective of the System Operation Chapter is to ensure co-ordinated operation of the IPS in order to achieve a high level of system reliability, safety and security.

2 SCOPE

This Chapter specifies the responsibilities of participants in relation to IPS operational issues such as—

- (a) reliability, security and safety;
- (b) co-ordination of planned equipment outages and maintenance.
- (c) operation of the IPS under normal and abnormal conditions;
- (d) ancillary services;
- (e) independent actions required and allowed by customers; and
- (f) Market operation actions required by the SO

3 RESPONSIBILITY

The SO shall be responsible for the safe and efficient operation of the IPS.

Participants shall set up operational procedures for their respective networks to ensure safe and efficient operation of the IPS.

International tie-line operations shall be governed by SAPP and other agreements.

4 SO RESPONSIBILITIES

The SO shall be responsible for the following:

4.1 *System reliability, safety and security*

- (a) The SO shall operate the IPS to achieve the highest degree of reliability practicable. Where there is a fault, remedial action shall be taken promptly to address any abnormal condition that may jeopardise the reliability and safety of the system.
- (b) The SO shall dispatch generation on the IPS subject to constraints of security, reliability and the environment.
- (c) The SO shall endeavour to retain international interconnections unless it becomes evident that continued parallel operation would jeopardise the integrity of the IPS.
- (d) Where it is unsafe to operate units in parallel with the system when critical levels of frequency and voltage result on the IPS from a disturbance, the SO shall ensure that separation or safe shut down of units shall be accomplished in such a way as to minimize the time required to resynchronise and restore the system to normal.
- (e) During or after a system disturbance, high priority shall be given to keeping all synchronised units running and connected to the IPS, or islanded on their own auxiliaries, in order to facilitate system restoration
- (f) Where there is loss of all generation and interconnections with neighbouring countries, blackstart services shall be contracted with at least two suitable facilities.
- (g) In the event of severe contingency, the SO shall ensure that instability, uncontrolled separation or cascading outages do not occur.
- (h) The SO shall be responsible for restoration of the TS after supply interruptions, in accordance with the restoration procedures for the IPS.

- (i) In the event of system disturbances, the SO shall endeavour to minimise adverse effects on customers.
- (j) The SO shall operate and maintain primary and backup control centers and associated facilities to ensure continuous operation of the IPS.

4.2 Operational measures

The SO shall establish and implement operating instructions, procedures and guidelines to cover the operation of the IPS under all system conditions. The SO shall maintain a database with version control of all such documents, and shall -

- (a) operate the IPS within its normal rating and defined technical standards.
- (b) manage constraints on the TS through the determination of operational limits and the purchase of the constrained generation ancillary services where applicable.
- (c) co-ordinate maintenance schedules of plant and equipment on the IPS and ensure that subsequent planned outages do not violate the reliability criteria of the IPS.
- (d) ensure adequate and reliable communication between all control centres, power stations and substations. Communications facilities to be provided and maintained by the customers are specified in the Information Exchange Chapter.
- (e) ensure the establishment of a common protection philosophy for the TS as required by the Network Chapter.
- (f) review protection settings on the IPS, after system alterations, and shall maintain an updated database of these settings.

5 SCHEDULING OF GENERATION AND ANCILLARY SERVICES

The SO shall —

- (a) develop daily twenty-four hours (24) day-ahead energy and ancillary services schedule before 14:00 each day; and
- (b) be responsible for executing and rescheduling of the energy and ancillary services schedule.

6 ANCILLARY SERVICES

The SO shall —

- (a) be responsible for the provision of all short-term reliability services for the IPS;
- (b) determine reliability targets for the purpose of acquiring ancillary services in consultation with affected participants. The reliability targets shall be selected so as to minimise the sum of the cost of providing the reliability plus the cost to the customer of limited reliability. The cost of providing suitable ancillary service levels shall be calculated annually for budgetary purposes; and
- (c) be responsible for procuring the required ancillary services needed to provide the required reliability.

7 OPERATIONAL AUTHORITY

The SO shall have the authority to issue operational instructions over the TS. Operational authority for other networks on the IPS shall lie with the respective Regional Operators.

Network control at the interface between the TNSP and a customer shall be in accordance with the operating agreements between the participants.

Except where otherwise stated in this section, a participant shall not be permitted to operate the equipment of another without the permission of such other participant. Where a participant wishes to operate another participant's equipment, the other participant has the right to test and authorise the relevant operating staff of the participant who wishes to operate the other participant's equipment in accordance with such other participant's standards before granting such permission.

Notwithstanding the provisions of Section 4, participants shall retain the right to safeguard the health of their equipment.

8 OPERATING PROCEDURES

The SO in liaison with relevant participants shall develop and maintain operating procedures for the safe, secure and reliable operation of the TS and for assets connected to the TS.

These operating procedures shall be issued to all participants.

All participants shall submit names of authorized personnel to the SO and Regional Operator.

Each customer shall comply with ZS418 Electrical Safety – Code of Practice and other relevant safety legislation and in doing so may develop in-house safety regulations.

The SAPP operating agreements shall apply in the case of operational liaison with all international power systems connected to the TS.

9 OPERATIONAL LIAISON

The SO shall sanction the shutting down and synchronising of units.

Where a participant experiences an emergency, the other participants shall assist that participant to the extent possible and as may be necessary, to ensure that it does not jeopardise the operation of the IPS or health of plant.

Whenever possible, distributors and end-use customers may be required to transfer load from one point of supply to another by performing switching operations on the network.

The TNSP, in consultation with a specific generator, shall compile a comprehensive maintenance, test and inspection plan for all equipment, systems and schemes installed in the specific HV yards.

The TNSP shall provide notification to generators of any work to be carried out on any protection, control and instrumentation circuits in adjacent HV yards. The TNSP shall compile re-commissioning programs for such work in consultation with the generator.

The TNSP and customers shall agree on the bus-bar configuration or configurations at each point of supply during normal and emergency conditions. Details of such configuration or configurations shall be included in the operating agreement between the participants.

Generators shall inform the SO of any environmental limitations that would affect the dispatch of the plant.

10 EMERGENCY AND CONTINGENCY PLANNING

The SO in conjunction with all participants shall develop, implement, maintain and ensure compliance with all contingency and emergency plans that are relevant to the performance of the IPS. These plans shall be consistent with Prudent Utility Practices.

Emergency plans shall allow for quick and orderly recovery from a partial or complete system collapse, with minimum impact on customer.

All emergency plans shall be periodically verified by actual tests to the greatest practical extent. In the event of such tests causing undue risk or undue cost to a participant, the SO shall take such risks or costs into consideration when deciding whether to conduct the tests. The cost of these tests shall be borne by the respective asset owners. The SO shall co-ordinate these tests with all affected participants.

The SO shall be responsible for determining all operational limits on the TS, updating them periodically and making them available to the participants.

The SO shall be responsible for keeping all participants informed of the status and expected status of the IPS and of the SAPP.

The SO, in conjunction with all Participants, shall conduct load flow studies annually.

11 SYSTEM FREQUENCY AND ACE CONTROL UNDER ABNORMAL FREQUENCY OR IMBALANCE CONDITIONS

The SO shall be responsible for the balancing of supply and demand in real time through the implementation of the energy schedules and utilisation of ancillary services.

11.1 *Description of normal frequency or balancing conditions*

The IPS is considered to be under normal frequency conditions when -

- (a) the immediate demand can be met with the available scheduled resources, including any contingency resources;
- (b) the frequency is within the range 49.85 to 50.15Hz; and
- (c) there are no security or safety contraventions.

11.2 *Operation during abnormal conditions*

When abnormal conditions occur, all relevant participants shall take corrective actions as stipulated in section 11 of this chapter. Appendix 12 gives guideline on operation during abnormal conditions.

The corrective action includes both supply-side and demand-side options. Where possible, warnings shall be issued by the SO on expected utilisation of any contingency resources.

The order in which emergency resources are to be used may change from time to time based on contractual arrangement. The SO shall issue an updated list of emergency resources available annually.

Termination of the use of emergency resources shall occur as the plant shortage situation improves and after frequency has returned to normal in the order "last in - first out"

During emergencies that require load shedding, the request to shed load shall be initiated in accordance with agreed procedures prepared and published by the SO.

Automatic under-frequency systems shall be kept armed at all times, apart from gas turbines, which shall be armed by the SO when a shortage is expected.

12 INDEPENDENT ACTION BY PARTICIPANTS

Each participant shall have the right to reduce or disconnect a point of connection under emergency conditions if such action is necessary for the protection of life or equipment. Advance notice of such action shall be given where possible and no financial penalties shall apply for such action.

Following such emergency operations as may be necessary to protect the integrity of the IPS or the safety of equipment and human life, the participants shall work diligently towards removing the cause of the emergency and the supply shall be reconnected immediately after the emergency conditions have passed.

13. VOLTAGE CONTROL

The SO shall be responsible for the voltage control of the TS, at transmission level voltages as well as at the interface between the TNSPs themselves and between the TNSPs and their customers.

TS voltages shall be controlled during normal operation to be within statutory limits at the points of supply and otherwise as agreed with customers.

The bilateral agreement for voltage and power factor control at the TNSP-TNSP boundary shall be submitted to the GCTC for approval.

14. FAULT REPORTING, ANALYSIS AND INCIDENT INVESTIGATION

An officer appointed by a participant shall report all incidents that materially affect the QOS on the IPS to the other participants.

14.1 *Generators, TNSPs, Distributors and End-use Customers*

Generators, TNSPs, distributors and end-use customers shall report verbally, any loss of output, tripping of units, change of status of AGC and governing, loss of transmission capacity or loss of major load to the SO within 15 minutes of the event occurring. This report will be prepared and sent by the control room staff on duty at the time of the incident.

Within seven days of a verbally reported incident, the participant that reported the incident shall submit a detailed written report to the SO in the form at Appendix 13.

Where a participant determines that a fault that had occurred on another participant's system requires further investigation, the participant shall issue an Incident Report in accordance with the Governance chapter.

14.2 *System Operator*

The SO shall investigate all incidents that materially affect the QOS on the IPS.

Such an investigation shall commence immediately after receipt of a preliminary report and all affected participants shall provide required information and participate in the investigation.

Participants shall, within mutually agreed time frames, implement recommended actions.

14.3 *Regional Operator*

The RO shall be responsible for the following—

- (a) retaining interconnections to the IPS;
- (b) restoration of the TS after supply interruptions;
- (c) maintaining its primary control centre;
- (d) determining TS operational limits;
- (e) coordinate maintenance schedules;
- (f) ensure reliable communication between its primary control centre and the SO;

- (g) ensure reliable communication between its control centres;
- (h) ensure establishment of TS protection philosophy;
- (i) maintaining TS protection database;
- (j) development, implementation and maintenance of operating procedures for its TS; and
- (k) ensuring compliance with emergency and contingency plans in harmony with the SO.

The RO shall—

- (a) operate the TS during normal and abnormal conditions;
- (b) schedule supply on the TS;
- (c) operate primary control centre;
- (d) manage TS constraints;
- (e) issue operations instructions on the TS;
- (f) verify commissioning programmes of equipment on TS;
- (g) authorize personnel to work on its TS; and
- (h) maintain record of bulk supplies.

14.4 *Root Cause and Forensic Analysis*

The GCTC will have the mandate to commission a team of experts to carry out root cause and forensic analyses of major incidents and faults on the system.

15 COMMISSIONING

The SO shall verify commissioning and maintenance programmes of the IPS to ensure co-ordination of activities.

Commissioning of new equipment associated with the transmission connection, or re-commissioning of such existing equipment, shall be agreed with the SO in writing.

The said aspects shall include, but not be limited to, the following:

- (a) commissioning procedures and programmes;
- (b) documents and drawings required;
- (c) proof of compliance with standards;
- (d) documentary proof of the completion of all required tests;
- (e) SCADA information to be available and tested before commissioning where such facilities are available; and
- (f) site responsibilities and authorities, etc.

Participants shall give a minimum notice period of one month, unless otherwise agreed, from the date of receipt of the request for all commissioning or re-commissioning. Where commissioning is likely to involve a requirement for dispatch or operating for test purposes, the participant concerned shall notify the SO of this requirement, including reasonable details as to the duration and type of the testing required.

When commissioning equipment at the point of connection, the TNSP shall liaise with the affected customers on all aspects that could potentially affect the customers' operation.

The TNSP and customers shall perform commissioning tests in order to confirm that TNSP and the customers' plant and equipment meet the requirements of this Code.

16 MAINTENANCE COORDINATION AND OUTAGE PLANNING

16.1 *Outage Management*

16.1.1 *Yearly Planned Maintenance Schedule*

Once all submissions regarding planned yearly maintenance works have been approved, they shall be consolidated into a yearly maintenance schedule, which shall be circulated to customers.

The service providers shall review the submissions taking into consideration system security, reliability, safety and commercial contracts, before making approval.

If the submissions have not been approved, the service providers shall request the customers affected to revise their plans and resubmit.

Once all the submissions have been approved, they shall be consolidated into a yearly maintenance schedule. Copies of the Yearly Maintenance Schedule shall be circulated to all customers.

Each TNSP shall submit their Yearly Maintenance Schedule to the SO for coordination and approval.

16.1.2 *Yearly Unplanned Outages*

If an unplanned outage allows time to give notice to the SO, this shall be done at least 2 weeks before the outage.

When the need for an unplanned outage is first identified it shall be communicated to the service providers as an unplanned outage indicating outage dates, times, reason, type of maintenance and urgency assigned to it.

At this point the service providers shall confirm the outage if it satisfies all the necessary requirements.

16.1.3 *Effecting of Outages*

The SO shall make available to customers an outage schedule of all planned outages on the TS. The outage schedule shall cover a period of one calendar year and shall indicate the status of the outage, that is to say, whether the outage is confirmed or not. The schedule shall be updated and republished monthly.

When an outage is cancelled or refused it is the responsibility of the person cancelling or refusing the outage to furnish reasons for cancellation or refusal, in writing. The person receiving the cancellation or refusal shall then enter this information into the system when changing the status to cancelled. This shall also apply to outages that are postponed.

16.2 *Emergency Outage*

In the case of emergency outages of customer equipment, customers shall inform service providers verbally of the need for the outage. However, within 24 hours after the outage has been effected, the outage request will need to be formalized in writing.

Service providers shall also inform customers in the case of emergency outages on their installations.

16.3 *Long Term Maintenance Planning for Generators*

Generators shall compile an indicative 5-year-ahead maintenance plan in consultation with the SO and TNSP. The generator shall then provide the SO with the following documents in the pro-forma format specified in the Information Exchange Chapter:

- (a) a 52-week-ahead outage plan per power station, showing planned outage and return dates and other known generation constraints, updated weekly by 15:00 hours every Thursday;
- (b) an annual maintenance or outage plan per power station, looking 5-years ahead, showing the same information as above and issued by 31 December of each year; and
- (c) a monthly variance report, explaining the differences between the above two reports.

Each generator shall ensure the absolute minimum deviation from its annual outage plan. Each deviation shall be negotiated with the SO and shall be accommodated as far as reasonably practical.

The SO shall coordinate network outages affecting unit output with related unit outages to the maximum possible extent.

The objectives to be used by the SO in this maintenance coordination are—

- (a) firstly maintaining adequate reserve levels at all times;
- (b) secondly ensuring reliability where TS constraints exist; and
- (c) thirdly maintaining acceptable and consistent real-time technical risk levels.

The SO shall provide forecasting of demand for the next day, the daily demand for the next seven days and the weekly demand for the next twelve months to the generators.

16.4 *Refusal or /cancellation of outages*

A participant shall not unreasonably refuse or cancel a confirmed outage. The direct costs related to the cancellation or postponement of an outage shall be borne by the participant cancelling or postponing the outage.

17 COMMUNICATION OF SYSTEM CONDITIONS, OPERATIONAL INFORMATION AND IPS PERFORMANCE

The SO shall communicate system conditions to all participants periodically.

The SO shall be responsible for providing participants with operational information as may be agreed from time-to-time and as specified in the Information Exchange Chapter. This shall include information regarding planned and forced outages on the IPS.

The SO shall annually publish expected fault levels, including the rupturing of relevant TS equipment, for each point of supply.

18 TELE-CONTROL

Where tele-control facilities are shared between the SO and other participants, the SO shall ensure that operating procedures are established in consultation with the participants.

CHAPTER 5 – INFORMATION EXCHANGE

1 OBJECTIVE

The objective of the Information Exchange Chapter is to define the obligations of parties with regard to the provision of information for the implementation of this Code in order to ensure the safe, reliable provision of transmission services.

2 SCOPE

Information requirements shall cover planning information, operational information and post-dispatch information.

The service-providers, the ERB and customers shall define the information requirements.

3 PRECEDENCE

In the event of inconsistencies between any part of this Code and the Information Exchange Chapter, with respect to information exchange, the provisions of the Information Exchange Chapter shall prevail.

4 INFORMATION EXCHANGE INTERFACE

For each type of information exchange, parties shall provide at the least the following information:

- (a) the name and contact details of the person(s) designated by the information owner to be responsible for provision of the information;
- (b) the names and contact details of and the parties represented by persons requesting the information; and
- (c) the purpose for which the information is required. This does not apply to Post-Dispatch Information.

The parties shall agree on appropriate confirmation procedures for the transfer of information.

5 SYSTEM PLANNING INFORMATION

5.1 *Objective*

The objectives of system planning information are to—

- (a) provide interaction of distributors' or end-use customers' planned system with the TS;
- (b) provide information for the TS to supply distributors and end-use-customers for system planning and development;
- (c) facilitate existing and planned connections; and
- (d) To provide fault condition information.

5.2 *Information required by TNSPs*

Customers may be required by a TNSP to provide information for reasons that include, but are not limited to—

- (a) planning and developing the TS;
- (b) monitoring current and future power system adequacy and performance; or
- (c) fulfil its statutory or regulatory obligations.

Customers shall provide the information required, without unreasonable delay and as the TNSP may request on a regular basis.

Customers shall submit to the TNSP the information listed in the following Appendices—

- (a) Appendix 14 – Distributor and end-use customer data; and
- (b) Appendix 15 – Generator Planning Data; and
- (c) Appendix 16 – Generator maintenance Data

Customers shall, upon request to upgrade an existing connection or when applying for a new connection, provide the TNSP with information as outlined in Appendix 17.

The TNSP may estimate any system planning information not provided by a customer as specified in Appendix 15 or 16. The TNSP shall take reasonable steps to reach agreement with the customer on estimated data items. The TNSP shall indicate to the customer all such estimated data items. The obligation to ensure the accuracy of the data remains with the customer.

Generators shall submit weekly updates to the relevant TNSP and SO all the maintenance planning information detailed in Appendix 16 with regard to each unit at each power station.

5.3 *Information required by Customers*

The TNSP shall provide customers or potential customers, upon request, with any relevant information that they require to properly plan and design their own networks or installations or comply with their obligations in terms of this Code.

The TNSP shall make available all the relevant information related to network planning as described in the Network Chapter.

5.4 *Information required by Generators*

The TNSP shall provide the generators with information about equipment and systems installed in the HV yards defined in Appendix 18.

The TNSP shall provide the generators with monthly rolling maintenance schedules for all planned work in HV yards for a period of one year in advance.

Log books on all vessels under pressure for receivers installed in HV yards shall be made available on request from the generator.

5.5 *Information required by the SO*

Participants shall provide the SO with updated technical data required for studying the behaviour of the IPS.

6 *OPERATIONAL INFORMATION*

6.1 *Pre-commissioning studies*

Participants shall meet all system planning information requirements before the commissioning test date, such as confirming any estimated values assumed for planning purposes or, where practicable, replacing them with validated actual values and with updated estimates for the future.

The SO shall perform pre-commissioning studies prior to sanctioning the final connection of new or modified plant. Such studies shall utilise data supplied by customers in accordance with section 5 of this Chapter, to verify that all control systems are correctly tuned and planning criteria have been satisfied.

The SO may request adjustments prior to commissioning should tuning adjustments be found to be necessary. The asset owner shall ensure that all system planning information records are maintained for reference for the operational life of the plant. Information shall be made available within reasonable time on request from the SO.

6.2 *General information requirements*

Measurements and indications to be supplied by participants to the SO shall be presented in such form as may be prescribed by the SO. These shall include, but not be limited to, the standards defined in Appendix 19.

Where required signals become unavailable or do not comply with applicable standards for reasons within the control of the provider of the information, such a participant shall report to the SO and restore or correct the signals or indications as soon as practical.

Following a modification to the TS, additional measurements or indications in relation to a participant's plant and equipment may be needed to meet TS requirements. In such an instance the SO shall notify the affected participants. The costs related to such modifications shall be on account of the participant effecting the modification.

On receipt of such notification from the SO the participant shall promptly ensure that such measurements or indications are made available at the RTU.

The data formats to be used and the fields of information to be supplied to the SO by the various participants shall be as defined in Appendix 19 of this code.

The TNSP shall provide feedback to customers regarding the status of equipment and systems installed in the substations where they are connected to the TS upon request. Agreements between parties shall determine the details and frequency of such feedback.

Plant status reports provided by the TNSP will also include contingency plans where applicable.

In the network where out-of-step relays are installed, the SO shall inform customers how the relays are expected to operate. The characteristics of such an islanded network shall be provided, based on the most probable local network configuration at such a time.

6.3 *Commissioning and notification*

Participants shall ensure that exciter, turbine governor and FACTS control system settings are implemented and are as finally recorded by the SO prior to commissioning.

Participants shall give the SO reasonable advance notice, as defined in the System Operation Chapter, of the time at which the commissioning tests will be carried out. The SO and the participant shall agree on the timeous provision of operational data as per Appendix 19.

Participants shall jointly notify the SO about the results of the verification of all measurements or indications for functionality and accuracy once every five years, so as to achieve overall accuracy of operational measurements within the limits agreed.

Commissioning records shall be maintained for reference by the SO for the operational life of the plant and shall be made available, within a reasonable time, to any participant upon request.

The asset owner shall communicate changes made to commissioned equipment, during an outage, to the SO and the relevant TNSP, before the equipment is returned to service. The TNSP shall keep commissioning records of operational data as per Appendix 19, for the operational life of the plant connected to the TS.

6.4 *Inter control centre communication*

This is communication between the SO and customer control centres. For normal and abnormal conditions, the following shall apply:

6.4.1 *Normal Conditions*

Control centres shall, upon request, provide each other with network information required for the security, safety, reliability and integrity of the TS. Such information exchange shall be electronic or paper-based within the time frame agreed upon by participants.

Participants shall optimise redundant control centre facilities where required in order to ensure the security of the TS and its safe operation.

6.4.2 Abnormal Conditions:

In abnormal conditions, communication shall be limited to the level necessary to enable the SO return the IPS to its normal condition. All control centres will therefore be required to await operational instructions from the SO, except in situations where a participant is islanded, in which case the participant would assume control of the participant's frequency and voltage, and await synchronising instructions from the SO. In exceptional situations, when a customer's network needs support from the IPS, they shall communicate the network status, and the kind of support needed.

(a) Communication facilities requirements

The communication facilities for voice and data that are to be installed and maintained between the SO and participants shall comply with the applicable IEC standards for SCADA and communications equipment.

(i) Tele-control

The information exchange shall support data acquisition from RTUs, SCSs and PCSs. The SO shall monitor the state of the IPS via telemetry from the RTU connected to the participants' plant.

The signals and indications required by the SO are defined in Appendix 17, together with such other information as the SO may, by notice to the participant reasonably require from time to time.

Participants shall interface via the standard digital interfaces, as specified by the SO. Interface cabinets shall be installed in the participants' plant and equipment room if required. The provision and maintenance of the wiring and signalling from the participants' plant and equipment to the interface cable shall be the responsibility of the participant.

The capability for the SO to deactivate and reactivate the scanning of a given RTU shall be provided by the participants, as shall the capability of monitoring the availability of all RTU centrally.

Participants shall comply with such tele-control requirements as may be applicable to the primary control centre and, as reasonably required, to the emergency control centre of the SO.

(ii) Telephone or facsimile

Each customer shall be responsible for the provision and maintenance of at least one telephone and one facsimile unit and these shall be reserved for operational purposes only. These facilities shall be continuously attended to and answered without undue delay.

The SO shall use a voice recorder for historical recording of all operational voice communication with participants. These records shall be available for at least three months after which will be disposed of. The SO shall make the voice records of an identified incident in dispute available within a reasonable time period after such a request from a participant or the ERB.

(iii) Electronic mail

The participants shall provide each other with electronic mailing addresses of contact persons as defined in this Code.

(b) SCADA and communication infrastructure at points of supply

(i) Access and security

The SO shall agree with participants the procedures governing security and access to the participants' SCADA, computer and other communication equipment. The procedures shall allow for adequate access to the equipment and information by the SO or its nominated representative for purposes of maintenance, repair, testing and the taking of readings.

Each participant shall designate a person with delegated authority to perform the duties of information owner in respect of the granting of access to information covered in this code to third parties, and shall disclose that person's name and contact details to the ERB. Each participant may, at its sole discretion, designate more than one person to perform these duties.

(ii) Time standards

All information exchange shall be GPS satellite time signal referenced. The SO shall ensure broadcasting of the standard time to relevant telecommunication devices in order to maintain time coherence.

(iii) Integrity of installation

The participant shall be responsible for optimising the reliability and security of the facilities to comply with the SO equipment and OEM minimum requirements. This includes the provision, at no charge to the SO, of an uninterruptible power supply with an eight-hour standby capacity.

(c) Data storage and archiving

The obligation for data storage and archiving shall lie with the information owner.

The systems must provide for clear and accessible audit trails on all relevant operational transactions. All requests that require an audit on a system shall be undertaken with reasonable notice to the parties.

The information owner shall keep all information for a period of at least five years, unless otherwise specified in this code, commencing from the date the information was created.

The parties shall ensure security against unauthorised access, use and loss of information.

Parties shall store planning information that is kept electronically for at least five years or for the life of the plant or equipment concerned, whichever is longer.

The SO shall store operational information in a historical repository sized for 3 years data. The data includes:

- (a)* TS time-tagged status information, change of status alarms and event messages;
- (b)* hourly scheduling and accounting information; and
- (c)* operator logs and operations.

An audit trail of all changes made to archive data should be maintained. This audit trail shall identify all changes made, and the time and date of the change. The audit trail shall include both before and after values of all content and structure changes.

7 POST-DISPATCH INFORMATION

7.1 *System information*

The SO shall provide participants, with at least the following system information:

- (a) hourly system total MW loading;
- (b) hourly individual power station MW sent out;
- (c) hourly system constraints and constrained generation;
- (d) hourly international tie-line power flow; and
- (e) system load flow data

7.2 *Generation information settlement*

Participants shall meet at least every three months to reconcile generation information. Should this information be classified as confidential, both parties shall treat it accordingly.

7.3 *Additional post-dispatch information*

The SO shall provide operational information regarding station dispatch and overall dispatch performance as specified in Appendix 20.

7.4 *Half-hourly demand metering data*

The TNSP shall provide participants with half-hourly metered data pertaining to their installations.

8 FILE TRANSFERS

The format of the files used for data transfer shall be agreed and defined by the supplier and receiver of the information.

The parties shall keep the agreed number of files for backup purposes so as to enable the recovery of information in the case of communication failures as shown in Appendix 21.

9 PERFORMANCE DATA

9.1 *Generator performance data*

Generators shall provide the SO with monthly-agreed performance indicators in relation to each unit at each power station as detailed in Appendix 22.

Generators shall report significant events, such as catastrophic failures, to the ERB within seven days of occurrence of such an event.

9.2 *Distributor and end-use customers performance*

The performance measurement of all distributors and end-use customers shall be in accordance with Zambian Standard ZS387, Electricity Supply – Power Quality and Reliability.

Distributors shall report periodic testing of under-frequency load shedding relays in the format given in Appendix 23.

9.3 *TNSP and SO Performance*

The TNSP and SO shall make available performance indicators listed in Appendix 24, monthly to the ERB and customers.

The TNSP shall provide to customers all performance indicators at each point of supply in accordance with Network Chapter.

9.4 *System Operation Performance Information*

The SO shall make at least the following IPS operational information available to all participants:

9.4.1 *Daily—*

- (a) the hourly actual demands of the previous day (MW); and
- (b) the reserve amounts over the morning, midday and evening peaks of the previous day (MW).

9.4.2 *Monthly—*

- (a) MW generated, Imports, Exports, available for distribution/sale and transmission losses;
- (b) generation Plan, Availability;
- (c) regulating reserve hours deficit over total hours;
- (d) number of frequency excursions;
- (e) report for each abnormal network condition and the action taken by the SO to restore normal operations; and
- (f) network constraints as reported by the TNSP.

9.4.3 *Annually—*

- (a) annual peak MW, date and hour;
- (b) annual minimum MW, date and hour; and
- (c) external Reliability Loss Factor (ERLF).

Following a TS disturbance, the TNSP shall make available all information collected via recorders installed at substations, to the SO for analyses within seven days of the date of the occurrence. Thereafter the SO shall compile a report that shall be available to affected customers on request.

10 *CONFIDENTIALITY OF INFORMATION*

Information exchanged between parties governed by this code shall be confidential, unless otherwise stated.

Confidential information shall not be transferred to a third party without the written consent of the information owner. Parties shall observe the proprietary rights of third parties for the purposes of this code. Access to confidential information within the organisations of parties shall be provided as reasonably required.

Parties receiving information shall use the information only for the purpose for which it was supplied.

The information owner may request the receiver of information to enter into a confidentiality agreement before confidential information is provided. A proforma agreement is included in Appendix 25 of this code.

The parties shall take all reasonable measures to control unauthorised access to information and to ensure secure information exchange. Parties shall report any leak of the information that is governed by a confidentiality agreement, as soon as practicable after it becomes aware to the leak, and shall provide the information owner with all reasonable assistance to ensure its recovery or destruction, as deemed appropriate by the information owner.

APPENDIX 1: AMENDMENT REQUEST FORM

GC A No:

Text to be amended: Section(s)

Page(s)

Change from (if additional space is required, please use attachments)

.....

.....

.....

Change to (if additional space is required, please use attachments)

.....

.....

.....

Reason for change(s)

.....

.....

.....

Change Initiated by..... Date

Checked by:

(Secretariat) Date

Approved/Not approved:

(GCTC CHAIRPERSON) Date

APPENDIX 2: EXEMPTION REQUEST FORM

GC E No:.....

Text to be exempted from: Section(s)..... Page(s).....

Reason for exemption (if additional space is required, please use attachments)

.....

.....

.....

.....

.....

Additional Information (if additional space is required, please use attachments)

.....

.....

.....

Duration:

From:.....To:.....

Exemption Applicant's Name:..... Date

Checked by:

(Secretariat)Date

Recommended by:

(GCTC Chairperson)Date

Approved/Not approved:

(ERB BOARD CHAIRPERSON)..... Date

APPENDIX 5: LOG OF DISPUTES BETWEEN PARTICIPANTS

[illegible]

APPENDIX 6: STANDARD APPLICATION FORM FOR *TRANSMISSION* CONNECTION

1.0 GENERAL

Company Name:
Company Registration No.:

Issued by Registrar of companies

Customer's Physical address

Customer's Physical Location

Postal Address

Customer's Preferred Postal address

Customer Contact Person	
Job Title:	
Name: (Initials & Surname)	
Telephone 1:	
Telephone 2/ Cell:	
Fax:	
Email address:	

Physical connection to the <i>Transmission</i> system required? (Y/N):
IF NOT INDICATE NATURE OF BUSINESS :

Application Date:	/ /
Customer's initial application date: dd – mm – yyyy	
Requested Completion Date	/ /
When customer wants supply available: dd – mm – yyyy	

Phasing if plant development is proposed in phases

Phase 1
Phase 2
Phase 3

dd - mm - yyyy

2. PLANT—ELECTRICAL

Desired Connection	
Voltage (kV):	
MVA:	

Capacity of Connection required (MVA)

Estimated Monthly		Declared	
Consumption:	MWh	Peak Load:	MW

Highest average no. of MW to be supplied in
any consecutive 30 minutes

Overall Plant	Limitations on
Power Factor:	Harmonics:

Emergency loading requirements if any:	MW
--	----

(For man winders, pumping and ventilation
in case of total loss of supply)

Short Circuit Levels:	MVA	MVA
-----------------------	-----	-----

State if there are any requirements for the
minimum fault level desired at *Point-Of-Supply*

Standard or Enhanced Reliability	
Connection:	

Temporary Connection:	Owner or Tenant:
-----------------------	------------------

If short term: period (*months*) for which
connection requiredCustomer owns/ rents property for this
application

3. ADDITIONAL INFORMATION

Point of Supply/Physical Connection Address:

Full description of the property/title deed where supply is required: Street address, lot no. etc
No Postal addresses

Usage category (please tick):

Industrial:	Commercial:	Distribution:
Generation:	International:	
Other (please specify):		

Nearest Existing Transmission System Connection:

Transmission substation closest to requested Point of supply

Other Transmission System Connections:

Does customer have other Transmission Connection Points of Supply

Special Instructions:

Customer's additional information regarding application

Note that further *information* may be required before a quote can be provided, as described
in the Grid Code

Cancel Date and Reason:

dd-mm-yyyy

4. TO BE FILLED IN BY SERVICE PROVIDER

Customer ID:			Existing Customer:		
Transmission-generated unique customer number					
Parent Customer ID:			Customer Type:		
Indicates if customer is subsidiary to existing customer			Indicate Individual/ Company /Partnership/ Other		
Application ID:			POS ID:		
Unique No. for current application			Unique No. applicable to Point-of-Supply		
Application Type ID					
NEW	INCR.	DECR.	CHANGE	LINES	
New POS/ Connection	Increase of Connection to:	Decrease of Connection to:	Change of Customer only	Existing Lines to be moved	
MVA	MVA	MVA	MVA	MVA	MVA
Size supply req.	Size supply req.	Size supply req.	Current supply	Current Supply	Size supply req.
Note: Only 1(one) Application Type ID per Application					
Priority Request			Priority Request		
Indicator:			Reason:		
High/ Medium/ Low			Motivation required for 'High' or 'Low' indicators		
Project ID:			Ref No/s		
			Details of any other transmission POS's linked to this customer		
Quotation Date: / /			Agreement Date: / /		
Date on which application completed dd-mm-yyyy			Date agreement completed dd-mm-yyyy		
Connection Fee Amount:			Connection Fee Payment Date: / /		
			Date agreement completed dd-mm-yyyy		
Connection Fee Receipt No.					
Unique No.					

APPENDIX 7: SURVEYING, MONITORING AND TESTING FOR GENERATORS

A.1 Introduction

This section specifies the procedures to be followed in carrying out the surveying, monitoring or testing to confirm the:

- (a) compliance by power stations with the ZAGC;
- (b) provision by power stations of ancillary services which they are required or have agreed to provide.

A.2 Scope

This code applies to *generators*.

A.3 Request for surveying, monitoring or testing

The *SO* may issue an instruction requiring a *power station* to carry out a test, at a time no sooner than 48 hours from the time of the instruction, to demonstrate that the relevant power station complies with the *GCRS*.

A.4 Ongoing Monitoring of a Unit's Performance

A *generator* shall monitor each of its *units* during normal service to confirm ongoing compliance with the applicable parts of this code. Any confirmed deviations detected must be reported to the *SO* within 2 working days.

A *generator* shall keep records relating to the compliance by each of its *units* with each section of this code applicable to that *unit*, setting out such *information* that the *SO* or *TNSP* reasonably requires for assessing power system performance under normal and abnormal conditions.

A *generator* shall provide to the *SO* a report, in January of each year, detailing the compliance by each of that *generator's* *units* with every section of the code during the past 12 *month* period. The template for this report is attached as an Appendix 3 in the *Information Exchange* Chapter.

A.5 Procedures

A.5.1. Unit Protection System - GCR 1	
APPLICABILITY AND FREQUENCY	
<i>Prototype study:</i> All new <i>power stations</i> coming on line or <i>power stations</i> where major refurbishment or upgrade of protection systems have taken place.	
<i>Routine review:</i> All <i>powerstations</i> every 5 to 6 years.	
PURPOSE	
To ensure that the relevant protection requirement in the <i>power station</i> is co-ordinated and aligned with the system requirements.	
PROCEDURE	
<i>Prototype:</i>	
1. Establish the System protection function and associated trip level requirements from <i>TNSP</i> and the records with the <i>SO</i> .	
2. Derive protection functions and settings that match the <i>power station</i> plant, <i>transmission</i> plant and system requirements.	
3. Confirm the stability of each protection function for all relevant system conditions.	
4. Document the details of the trip levels, stability calculations for each protection function.	
5. Convert protection-tripping levels for each protection function into per <i>unit</i> base.	
6. Consolidate all settings in per <i>unit</i> base for all protection functions in one document	
7. Derive actual relay dial setting details and document the relay setting sheet for all protection functions.	
8. Document the position of each protection function on one single line diagram of the generating <i>unit</i> and associated connections.	
9. Document the tripping functions for each tripping function on one tripping logic diagram	
10. Consolidate detail setting calculations, per <i>unit</i> setting sheets, relay setting sheets, plant base <i>information</i> the settings are based on, tripping logic diagram, protection function single line diagram and relevant protection relay manufacturers <i>information</i> into one document.	
11. Submit to <i>TNSP</i> and <i>SO</i> for their acceptance and update.	
12. Provide <i>TNSP</i> and <i>SO</i> with one original reference copy and one working copy.	
<i>Review:</i>	
13. Review Items 1 to 10 above.	
14. Submit to <i>TNSP</i> and <i>SO</i> for their acceptance and update.	
15. Provide <i>TNSP</i> and <i>SO</i> with one original reference copy and one working copy.	
ACCEPTANCE CRITERIA	
All protection functions are set to meet the necessary protection requirements of the <i>transmission network</i> and <i>power station</i> plant with minimal margin.	
Optimal fault clearing times, and maximum plant availability	
Submit a report to <i>TNSP</i> and <i>SO</i> one <i>month</i> after commissioning for prototype study or 5 to 6 yearly for routine tests.	

Protection Integrity Tests	APPLICABILITY <i>Prototype test:</i> All new <i>power stations</i> coming on line and all other <i>power stations</i> after major modifications or refurbishment of protection or related plant. <i>Routine test:</i> All <i>Power Stations</i> 5 to 6 yearly or after major overhaul of plant. PURPOSE To confirm that the protection has been wired and function according to the specified. PROCEDURE (a) Apply final settings as per agreed documentation to all protection functions. (b) With the generator unit off load and de-energized, inject appropriate signals into every protection function and confirm correct operation and correct calibration. Document all protection function operations. (c) Carry out trip testing of all protection functions, from origin (e.g. Buchholz relay) to all tripping output devices (e.g. HV Breaker). Document all trip test responses. (d) Apply short circuits at all relevant protection zones and with generator at nominal speed excite generator slowly, record currents at all relevant protection functions, and confirm correct operation of all relevant protection functions. Document all readings and responses. Remove all short circuits. With the generator at nominal speed, excite generator slowly recording voltages on all relevant protection functions. Confirm correct operation and correct calibration of all protection functions. Document all readings and responses. ACCEPTANCE CRITERIA All protection functions fully operational and operate to required levels within the relay OEM allowable tolerances. Measuring instrumentation used shall be sufficiently accurate and calibrated to traceable standard. Submit a report to <i>TNSP</i> and <i>SO</i> one <i>month</i> after test.
----------------------------	---

A.5.2 Excitation System Grid Code Requirement GCR 2	
Excitation and Setting Integrity Study	Parameter/APPLICABILITY AND FREQUENCY <i>Prototype study:</i> All new <i>power stations</i> coming on line or <i>power stations</i> where major refurbishment or upgrade of protection systems have taken place. <i>Routine review:</i> All <i>power stations</i> every 5 to 6 years. PURPOSE To ensure that the Excitation systems in the <i>power stations</i> is co-ordinated and aligned with the system requirements. PROCEDURE <i>Prototype:</i> 1. Establish the System excitation system performance requirements from <i>TNSP</i> and <i>SO</i>. 2. Derive a suitable model for the excitation system according to <i>IEEE421.5</i> or <i>IEC 60034.16.2</i>. Where necessary, nonstandard models (non <i>IEC</i> or <i>IEEE</i>) shall be created. This may require <i>frequency</i> response and bode plot tests on the excitation system as described in <i>IEEE 421.2.1990</i>. 3. Submit the model to <i>TNSP</i> and <i>SO</i> for their acceptance. 4. Derive excitation system settings that match the <i>power station</i> plant, <i>transmission</i> plant and system requirements. This includes the settings of all parts of the excitation system such as the chop-over limits and levels, limiters, protection devices, alarms. 5. Confirm the stability of the excitation system for relevant excitation system operating conditions system conditions. 6. Document the details of the trip levels, stability calculations for each setting and function. 7. Convert settings for each function into per <i>unit</i> base and produce a high-level dynamic performance model with actual settings in p.u. values. 8. Derive actual card setting details and document the relay setting sheet for all setting functions.

	9. Produce a single line diagram / block diagram of all the functions in the excitation system and indicate signal source. 10. Document the tripping functions for each tripping on one tripping logic diagram. 11. Consolidate detail setting calculations, model, per <i>unit</i> setting sheets, relay setting sheets, plant base <i>information</i> the settings are based on, tripping logic diagram, protection function single line diagram and relevant protection relay manufacturers <i>information</i> into one document. 12. Submit to <i>TNSP</i> and <i>SO</i> for their acceptance and update by providing them with one original master copy and one working copy. <i>Review:</i> Review Items 1 to 10 above. Submit to <i>Transmission</i> for their acceptance and update. Provide <i>Transmission</i> with one original master copy and one working copy update if applicable. ACCEPTANCE CRITERIA Excitation system is set to meet the necessary control requirements in an optimized manner for the performance of the <i>transmission</i> and <i>power station</i> plant. Excitation system operates stable both internally and on the network.
--	---

Parameter	
Excitation Response Tests	APPLICABILITY <i>Prototype test:</i> All new <i>power stations</i> coming on line and all other <i>power stations</i> after major modifications or refurbishment of protection or related plant. <i>Routine test:</i> All <i>Power Stations</i> 5 to 6 yearly or after major overhaul of plant. <i>Prototype test:</i> All new <i>power stations</i> coming on line and all other <i>power stations</i> after major modifications or refurbishment of protection or related plant. <i>Routine test:</i> All <i>Power Stations</i> 5 to 6 yearly or after major overhaul of plant. PURPOSE To confirm that the excitation system performs as per designed and installed. PROCEDURE With the generator off line, carry out <i>frequency</i> scans / bode plot tests on all circuits in the excitation system critical to the performance of the excitation system. With the generator in the open circuit mode, carry out the Large signal performance testing as described in <i>IEEE 421.2</i> of 1990. Determine Time response, Ceiling voltage, voltage response. With the <i>generator</i> connected to the network and loaded, carry out the small signal performance tests according to <i>IEEE 421.2</i> 1990. Also carry out power system stabiliser tests and determine damping with and without Power System stabiliser. Document all responses. ACCEPTANCE CRITERIA Excitation system meets the necessary control requirements in an optimised manner for the performance of the <i>transmission</i> and <i>power station</i> plant as specified. Excitation system operates stable both internally and on the network. Power System stabilisers set for optimised damping.

Parameter	APPLICABILITY
Reactive Power Capability	Prototype test: All new <i>power stations</i> coming on line and all other <i>power stations</i> after major modifications or refurbishment of protection or related plant. PURPOSE To confirm that the reactive Power Capability specified are met. PROCEDURE The duration of the test will be for a period of up to 60 minutes during which period the System voltage at the Grid Entry Point for the relevant Generating <i>Unit</i> will be maintained by the <i>Generator</i> at the voltage specified by adjustment of Reactive Power on the remaining Generating <i>Units</i>, if necessary. ACCEPTANCE CRITERIA Generating <i>Unit</i> will pass the test if it is within $\pm 5\%$ of the capability registered with <i>TNSP</i> and <i>SO</i>. Submit a report to <i>TNSP</i> and <i>SO</i> one month after test.
Intermittent Generation	Intermittent generation is defined as generation that is not dispatched by the system operator and is not committed to the system operator. Intermittent generation is defined as generation that is not dispatched by the system operator and is not committed to the system operator.

A.5.4 Governing System GCR 4.

Parameter	APPLICABILITY
<i>Governing Response Tests</i>	<i>Prototype test:</i> All new <i>power stations</i> coming on line and all other <i>power stations</i> after major modifications or refurbishment of protection or related plant. <i>Routine test:</i> All <i>Units</i> to be monitored continuously, additional tests may be requested by the <i>SO</i>. PURPOSE Prove the <i>unit</i> is capable of the minimum requirements required for <i>Governing</i> PROCEDURE (a) <i>Frequency</i> or speed deviation to be injected on the <i>Unit</i> for 10 minutes. (b) Real Power Output of the <i>Unit</i> is to be measured and recorded. ACCEPTANCE CRITERIA Minimum requirements of the <i>ZAGC</i> are met

A.5.5 Black Start Capability GCR 5

Parameter	APPLICABILITY
Black Starting	<i>Routine Test: Power stations that have contracted under the ancillary services to supply station Black start services. When called for by SO but not more than once every 2 years</i> PURPOSE Demonstrate that a power station has Black Start Capability PROCEDURE The relevant Generating Unit shall be Synchronised and Loaded: 1. All the Auxiliary generators in the station in which that Generating Unit is situated, shall be Shutdown. 2. The Generating Unit shall be De-Loaded and De-Synchronised and all AC electrical supplies to its Auxiliaries shall be disconnected. 3. The station shall then execute its Blackstart Procedure and restart the relevant Generating Unit. Synchronise it to the System and wait for loading instructions to be given by SO. All Black Start Tests shall be carried out at the time agreed with the SO and shall be undertaken in the presence of TNSP and SO representatives, who shall be given access to all information relevant to the Black Start Test. ACCEPTANCE CRITERIA A Black Start Station shall fail a Black Start Test if the Black Start Test shows That it does not have a Black Start Capability (i.e. if the relevant Generating Unit Fails to be Synchronised to the System within thirty (30) minutes of the commencement of the Blackstart procedure. A written report shall be submitted to the SO and TNSP one month after test.

APPENDIX 8: TRANSMISSION DRAWINGS SYMBOL SET AND LAYOUT
CONVENTIONS

IEC 60617 "Graphical symbols for diagrams" or an appropriate *Zambian* standard shall be used as the standard for electrical drawings, unless otherwise agreed between a service provider and a *customer*

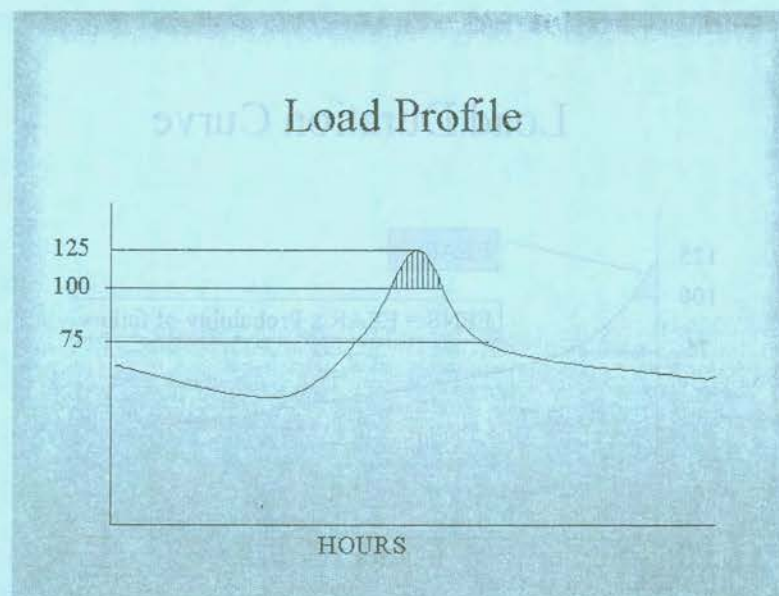
SYMBOL	DESCRIPTION
2=	CONDUCTOR
2=	CONDUCTOR

APPENDIX 9: TECHNICAL VOLTAGE LIMITS

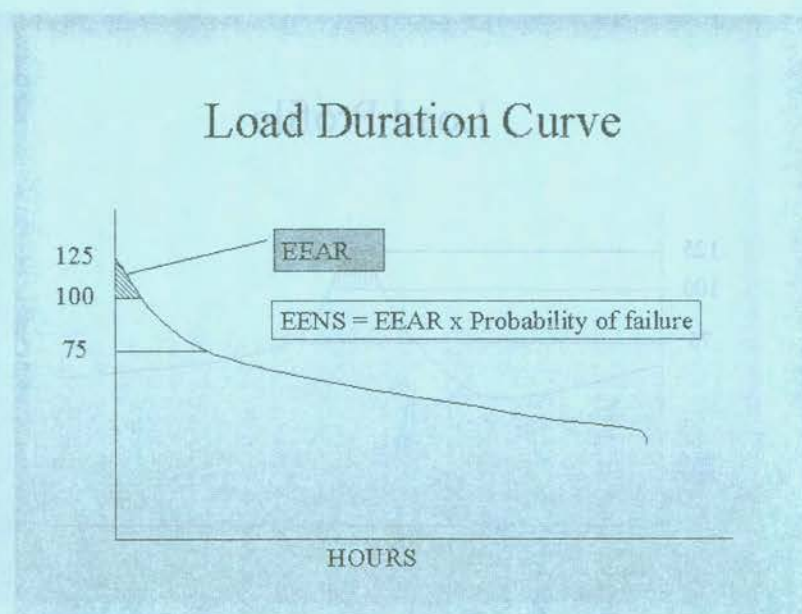
Voltage level V	Compatibility level %
< 88000	± 10
≥ 88000	± 5

Note — In the absence of any agreement to the contrary, the supply voltage shall not deviate from the compatibility level for any period longer than 10 consecutive minutes.

APPENDIX 10: TYPICAL LOAD PROFILE



APPENDIX 11: LOAD DURATION CURVE



APPENDIX 12: OPERATION DURING ABNORMAL CONDITIONS

APPENDIX 12: OPERATION DURING ABNORMAL CONDITIONS

CONDITION FOR USAGE	RESOURCES IN DEFAULT ORDER OF USAGE
Warnings	
WHEN A SHORTFALL IN CAPACITY IS EXPECTED TO OCCUR, ISSUE WARNINGS UNTIL SUFFICIENT CAPACITY IS OBTAINED TO COVER THE SHORTFALL When there is an appreciable loss of load leading to increased frequency, issue warnings until load has been restored.	(a) Emergency level generation warning (b) load shedding warning (c) Emergency generation reduction (d) Generation plant shut down
Gradual frequency decline	
IF THE FREQUENCY FALLS BELOW 50HZ AND ABNORMAL VOLTAGE CONDITION EXISTS	(e) Run unscheduled generation units (f) Interruptible load shedding (g) Declare emergency to other SAPP Control Areas
Rapid Frequency Decline	
ENSURE OPERATION OF UNDER-FREQUENCY PROTECTION SCHEMS	(h) Under-frequency relay protection schemes

APPENDIX 13: FORMAT FOR PRELIMINARY INCIDENT REPORT

- (a) Time of the incident
- (b) Location
- (c) Plant and/or apparatus directly involved (and not merely affected by the incident)
- (d) Description of the incident
- (e) Demand (MW) and/or generation (MW) interrupted and duration of interruption
- (f) Generating unit – frequency response (MW and Hz correction achieved subsequent to the incident)
- (g) Generating Unit– MVA_r performance (change in output subsequent to the incident)
- (h) Change in tie-line flow, where applicable
- (i) Estimated time and date of return to service
- (j) Measures taken to prevent/ mitigate recurrence

APPENDIX 14: DISTRIBUTOR AND END-USE CUSTOMER DATA

Unless otherwise indicated, the following information shall be supplied to the TNSP, prior to connection and then updated as and when changes occur.

(a) Demand data

Connection capacity	Connection capacity required (MW)
Measured and forecast data (annually)	For each point of supply, the information required is as follows: (i) A 5 and 15-year demand forecast. (See Appendix 9). (ii) A description setting out the basis for the forecast. (iii) The season of peak demand. (iv) Quantification of the estimated impact of embedded generation where applicable. (See Appendix 9)
User network data	(i) Electrical single-line diagram of user network to a level of detail to be agreed with the service providers, including the electrical characteristics of circuits and equipment (R1, X1, B1, R0, X0, B0, continuous and probabilistic ratings). (ii) Contribution from customer network to a three-phase short circuit at point of connection (including that from embedded generation if available) (iii) Connection details of all customer transformers, shunt capacitors, shunt reactors etc. connected to the secondary voltage levels of the customer connected the TS, (the requirement here is for data pertaining to the network connecting shunt capacitors, harmonic filters, reactors, SVCs, etc. to the point of supply for purposes of conducting harmonic resonance studies.) (iv) Electrical characteristics of all circuits and equipment at a voltage lower than secondary voltage levels of the customer connected the TS that may form a closed tie between two connection points on the TS.
Standby supply data (annually)	The following information is required from each distributor and end-use customer that can take supply from more than one supply point: (i) Source of standby supply (alternative supply point(s)) (ii) Standby capacity required (MW)
General information	For each new connection from a distributor or end-use customer, the following information is required: (i) Number and type of switch-bays required (ii) Load build-up curve (in the case of new end-user plant) (iii) Supply date (start of load build-up) (iv) Temporary construction supply requirements (v) Load type (e.g. arc furnaces, rectifiers, rolling mills, residential, commercial, etc.) (vi) Annual load factor (vii) Power factor (including details of harmonic filters and power factor correction capacitors) (viii) Special requirements (e.g. QOS) (ix) Other information reasonably required by the service-providers to provide the customer with an appropriate supply (e.g. pollution emission levels for insulation design)
Disturbing loads	Description of any load on the power system that could adversely affect the SO target conditions for power quality and the variation in the power quality that can be expected at the point connected to the TS. (The areas of concern here are, firstly, motors with starting currents referred back to the nominal voltage at the point of supply exceeding 5% of the fault level at the point of supply and secondly, arc furnaces likely to produce flicker levels at the point of supply in excess of the limits specified in ZS387 Electricity Supply - Power Quality and Reliability standard. The size limit for arc furnaces is subject to local conditions, in respect of fault level at the point of supply and background flicker produced by other arc furnaces and other equipment that will produce harmonics and/or negative and zero sequence current components, such as large AC/DC rectification installations.)

(a) Transmission system connected transformer data

	Symbol	Units
Number of windings		
Vector group		
Rated current of each winding		A
Transformer rating		MVA _{rated}
Transformer tertiary rating		MVA
Transformer nominal LV voltage		kV
Transformer nominal tertiary voltage		kV
Transformer nominal HV voltage		kV
Tapped winding		HV/MV/LV/None ¹
Transformer ratio at all transformer taps		
Transformer impedance (resistance R and reactance X) at all taps	$R+jX$	% on rating MVA _{rated}
For three-winding transformers, where there are external connections to all three windings, the impedance (resistance R and reactance X) between each pair of windings is required, measured with the third set of terminals open-circuit.	$Z_{HV/MV}$	% on rating MVA _{rated}
	$Z_{HV/LV}$	% on rating MVA _{rated}
	$Z_{MV/LV}$	% on rating MVA _{rated}
Transformer zero sequence impedances at nominal tap		
Zero phase sequence impedance measured between the HV terminals (shorted) and the neutral terminal, with the LV terminals open-circuit.	Z_{H0}	Ohm
Zero phase sequence impedance measured between the shorted HV terminals and the neutral terminal, with the LV terminals short-circuited to the neutral.	Z_{H0}	Ohm
Zero phase sequence impedance measured between the LV terminals (shorted) and the neutral terminal, with the HV terminals open-circuit.	Z_{L0}	Ohm
Zero phase sequence impedance measured between the LV terminals (shorted) and the neutral terminal, with the HV terminals short-circuited to the neutral.	Z_{L0}	Ohm
Zero phase sequence leakage impedance measured between the HV terminals (shorted) and the LV terminals (shorted), with the Delta winding closed.	Z_{00}	Ohm
Earthing arrangement, including LV neutral earthing resistance and reactance core construction (number of limbs, shell or core type)		
Open circuit characteristic		Graph

Transformer test certificates, from which actual technical detail can be extracted as required, are to be supplied on request.

¹ Delete what is not applicable

(c) Shunt capacitor or reactor data requirements

For each shunt capacitor or reactor or power factor correction equipment or harmonic filters with a rating in excess of 10MVar connected to or capable of being connected to a customer network, the customer shall inform and provide the TNSP with the specific shunt capacitor or reactor data as well as network details necessary to perform primarily harmonic resonance studies. The customer shall inform the TNSP of his intention to extend or modify this equipment.

If any participant finds that a capacitor bank of 10MVar or less is likely to cause harmonic resonance problems on the TS, he shall inform the TNSP. The 10MVar minimum size limit shall thereafter be waived in respect of the affected network for information reporting purposes in respect of this code, and the TNSP shall inform the affected participants of this fact and request the additional data. If the affected network is modified or reinforced to the extent that capacitor banks of 10MVar or less no longer cause harmonic resonance problems on the TS, the TNSP shall inform the affected participants that information reporting requirements have returned to normal.

Any party to this code investigating a complaint about harmonic distortion, shall have the right to request such additional information (including, but not restricted to, data from harmonic distortion measuring devices) from parties in the vicinity of the source of the complaint as may reasonably be required to complete the investigation.

Shunt capacitor or reactor rating	Rating (MVar)
Reactor/capacitor/harmonic filter ¹	
Location (station name)	
Voltage rating	kV
Resistance/reactance/susceptance ² of all components of the capacitor or reactor bank	
Fixed or switched	
If switched	Control details (manual, time, load, voltage, etc.)
If automatic control	Details of settings. If under FACTS device control (e.g. SVC), which device?

(d) Series capacitor or reactor data requirements

Series capacitors are installed in long transmission lines to increase load transfer capability.

Series reactors are installed to limit fault levels, or to balance load sharing between circuits operated in parallel that would otherwise not share load equitably, or to balance load sharing on an interconnected network.

Reactor/capacitor ²	
LOCATION (SPECIFY SUBSTATION BAY WHERE APPLICABLE)	
Voltage rating	kV
Impedance rating	Ohm or MVar
Current rating (continuous and emergency, maximum times for emergency ratings)	Continuous: A Hours A Hours A Hours A

Note: if a series capacitor or reactor is located in a dedicated reactor or capacitor station (i.e. a substation built to hold only the series reactor or capacitor), the lines or cables linking it to each remote end substation must be specified as separate circuits under line or cable data.

1. FACTS DEVICES

FACTS devices enable system parameters (voltage, current, power flow) to be accurately controlled in real time. Because of their cost, they are generally used only if cheaper, more conventional, solutions cannot deliver the required functionality.

Applications requiring rapid control capability include the following:

- (a) Voltage regulation following loss of a system component, generation, large load;
- (b) Arc furnace voltage flicker mitigation;
- (c) Negative phase sequence voltage compensation;
- (d) SSR damping;
- (e) Machine transient stability enhancement;
- (f) System load transfer capability enhancement;
- (g) Load sharing control in interconnected, deregulated, networks.

The most commonly used FACTS device is the SVC. Other FACTS devices made possible by advances in power electronics and control systems include STATCON, TCSC, thyristor controlled tap changer, thyristor controlled phase shifter, BES (Battery Energy Storage), and UPC. The common factor is rapid control capability.

Because FACTS devices are purpose-designed for their specific applications, the following data is required:

Name	Station, HV voltage, device number
Type	(SVC, STATCON, TCSC, etc.)
Configuration: Provide a single line diagram showing all HV components and their MVA/MVAr and voltage ratings, with all controlled components identified as such.	
Control system: Provide a block diagram of the control system suitable for dynamics modeling.	
Primary control mode	Voltage control, arc furnace flicker mitigation, negative phase sequence voltage control, etc.

Customers are required to perform, or cause to be performed, harmonic studies to ensure that their installation does not excite harmonic resonance, and that harmonic distortion levels at the PCC with the TS do not exceed the limits specified in ZS 387 Electricity Supply - Quality and Reliability standard.

2. HVDC

HVDC is a form of FACTS device because of the rapid control capabilities. However, HVDC is treated separately because its primary function is the transmission of real power.

HVDC is used to connect two systems that are not necessarily interconnected via the AC network (and thus in synchronism), or even at the same nominal frequency.

Customers wishing to connect HVDC systems to the TS shall supply a single line diagram showing all HV plant (including valve bridges) forming part of the HVDC system, plus additional HV plant required for its proper operation, e.g. harmonic filters, synchronous condensers, FACTS devices, etc. Customers and the TNSP shall cooperate in performing, or causing to be performed, studies to determine network-strengthening requirements needed to accommodate the HVDC system without violating the planning criteria specified in the Network Chapter. In addition, customers shall thereafter perform, or cause to be performed, studies to demonstrate that the proposed HVDC system does not exceed QOS parameters specified in ZS387 Electricity Supply - Power Quality and Reliability standard, and where applicable shall specify what additional HV plant will be required to ensure compliance with this standard.

(e) Information on customer networks

If a customer will have two or more points of supply from the TS, including the one applied for, the customer shall specify the amount of load to be transferred from existing points of supply to the new one under normal conditions as well as under contingencies. The same requirement applies to any embedded generators within the customer's network, since they affect fault levels as well as net load on the system.

The customer shall also specify whether he intends to interconnect two or more transmission points of supply via his network. In such circumstances the customer shall provide detailed information on the lines and cables used.

Where a circuit consists of two or more segments of different characteristics (different overhead line tower and/or conductor bundle types and/or underground cable types), each section shall be specified separately.

	Units
Line description	Name ("from" busbar, "to" busbar, circuit number and, where applicable, line section number numbered from the "from" bus end)
Line voltage (specify separately for dual voltage multi-circuit lines)	kV
Single / double / multiple circuit	
Standard suspension tower information (to confirm impedance): Supply copy of tower drawing, or sketch drawing showing co-ordinates of shield wire and phase conductor bundle attachment points relative to tower centre line and ground level at nominal tower height.	
Phase sub-conductor type (per circuit)	
Number of sub-conductors per phase conductor bundle	
Sub-conductor spacing, if applicable (supply sketch showing phase conductor bundle geometry and attachment point)	Mm
Number of earth wires	
Earthwire description	
Line length	Km
Conductor parameters (R , X , B , R_0 , X_0 , B_0) (specify)	Ohmic values or p.u. on 100MVA base
Conductor normal and emergency ratings	Ampere or 3 phase MVA at nominal voltage

Cable data

Cable description circuit number, and where	Name ("from" busbar, "to" busbar, applicable, line section number numbered from the "from" bus end)
Voltage rating	kV
Type (copper/aluminium)	(Delete what is not applicable)
Size	mm ²
Impedance (R , X , B , R_0 , X_0 , B_0)	Ohms or p.u. on 100MVA base (specify)
Length	Km
Continuous and (where applicable) emergency current rating and time limit	Amp or MVA at nominal voltage (specify), hours maximum at emergency rating

APPENDIX 15: GENERATOR PLANNING DATA

Unless otherwise indicated, the following information shall be provided to the TNSP and SO, prior to connection and then updated as and when changes occur.

(a) Power station data

Generator name (Owner)	
Power station name	
Number of units	
Primary fuel type / prime mover	For example gas, hydro, fossil or nuclear
Secondary fuel type	For example oil
Capacity requirement (MW)	Generation sent-out connection capacity required
Blackstart Capability	CONTINGENCY PLANS TO RESTART THE GENERATORS' OWN SYSTEMS AFTER A TOTAL OR PARTIAL COLLAPSE IN THE NETWORK.
Partial load rejection capability	A description of the amount of load the unit can automatically govern back, without any restrictions, as a function of the load at the point of governing initiation.

(b) Unit data

Unit number	
Capacity	Unit capacity (MW)
Normal maximum continuous generation capacity	MW
Normal maximum continuous sent out capacity	MW
Unit auxiliary active load	MW
Unit auxiliary reactive load	MVA _r
Maximum (EL1) generating capacity	MW
Maximum (EL2) sent out capacity	MW
Normal minimum continuous generating capacity	MW
Normal minimum continuous sent out capacity	MW
Generator rating (Mbase)	MVA
Normal maximum lagging power factor	MVA _r
Normal maximum leading power factor	MVA _r
Governor droop	
Forbidden loading zones	MW
Terminal voltage adjustment range	KV
Short circuit ratio	
Rated stator current	Amp
Time to synchronise with penstock charged (warm)	Hour
Time to synchronise with penstock empty from (cold)	Hour
Normal loading rate	MW/min
Normal de-loading rate	MW/min
Partial load rejection capability	% MW name plate rating

Description	Data
Capability chart showing full range of operating capability of the generator, including thermal and excitation limits	Diagram
Open circuit magnetisation curves	Graph
Short circuit characteristic	Graph
Zero power factor curve	Graph
V curves	Diagram

Documents	Description
Protection setting document	A document agreed and signed by the SO containing the following: (i) A section defining the base values and per unit values to be used (ii) A single line diagram showing all the protection functions and sources of current and voltage signals (iii) A protection tripping diagram(s) showing all the protection functions and associated tripping logic and tripping functions (iv) A detailed description of setting calculation for each protection setting, discussion on protection function stability calculations, and detailed dial settings on the protection relay in order to achieve the required setting (v) A section containing a summary of all protection settings on a per unit basis (vi) A section containing a summary for each of the protection relay dial settings/programming details (vii) An annex containing plant information data (e.g. OEM data) on which the settings are based (viii) An annex containing OEM information sheets or documents describing how the protection relays function
Excitation setting document	A document agreed and signed by the SO containing the following: (i) A section defining the base values and per unit values to be used (ii) A single line diagram showing all the excitation system functions and all the related protection tripping functions (iii) An excitation system transfer function block diagram in accordance with IEC standard models (iv) A detailed description of setting calculation for each of the excitation system functions, discussion on function stability calculations, and detailed dial settings on the excitation system in order to achieve the required setting (v) A section containing a summary of all settings on a per unit basis (vi) A section containing a summary for each of the excitation system dial settings/programming details (vii) An annex containing plant information data (e.g. OEM data) on which the settings are based (viii) An annex containing OEM information sheets or documents describing the performance of the overall excitation system and each excitation function for which a setting is derived
Governor setting document	A document agreed and signed by the SO containing the following: (i) A section defining the base values and per unit values to be used (ii) A single line diagram showing all the excitation system functions and all the related protection tripping functions (iii) An excitation system transfer function block diagram in accordance with IEC standard models (iv) A detailed description of setting calculation for each of the governor system functions, discussion on function stability calculations, and detailed dial settings on the governor system in order to achieve the required setting (v) A section containing a summary of all settings on a per unit basis (vi) A section containing a summary for each of the governor system dial settings/programming details (vii) An annex containing plant information data (e.g. OEM data) on which the settings are based (viii) An annex containing OEM information sheets or documents describing the performance of the overall governor system and each governor function for which a setting is derived

(c) Reserve capability

The generator shall provide the SO with the reserve capability of each power station. The reserve capability shall be indicated as defined in the SAPP Operating Guidelines (1996)

(d) Unit parameters

	Symbol	Units
Direct axis synchronous reactance	X_d	% on rating
Direct axis transient reactance saturated	$X'_{d, sat}$	% on rating
Direct axis transient reactance unsaturated	$X'_{d, unsat}$	% on rating
Sub-transient reactance unsaturated	$X''_d = X''_q$	% on rating
Quad axis synchronous reactance	X_q	% on rating
Quad axis transient reactance unsaturated	$X'_{q, unsat}$	% on rating
Negative phase sequence synchronous reactance	X_2	% on rating
Zero phase sequence reactance	X_{0q}	% on rating
Turbine generator inertia constant for entire rotating mass	H	MW s/MVA
Stator resistance	R_a	% on rating
Stator leakage reactance	X_l	% on rating
Poiter reactance	X_p	% on rating
Generator time constants:		
(i) Direct axis open-circuit transient	T_{do}'	sec
(ii) Direct axis open-circuit sub-transient	T_{do}''	sec
(iii) Quad axis open-circuit transient	T_{qo}'	sec
(iv) Quad axis open-circuit sub-transient	T_{qo}''	sec
(v) Direct axis short-circuit transient	T_d'	sec
(vi) Direct axis short-circuit sub-transient	T_d''	sec
(vii) Quad axis short-circuit transient	T_q'	sec
(viii) Quad axis short-circuit sub-transient	T_q''	sec
Speed damping	D	
Saturation ratio at 1 pu terminal voltage	S(1.0)	
Saturation ratio at 1.2 pu terminal voltage	S(1.2)	

e) Excitation system

The generator shall fill in the following parameters or supply a Laplace-domain control block diagram in accordance with IEEE or IEC standard excitation models (or as otherwise agreed with the SO) completely specifying all time constants and gains to fully explain the transfer function from the compensator or unit terminal voltage and field current to unit field voltage. Customers shall perform, or cause to be performed, small signal dynamic studies to ensure that the proposed excitation system and turbine governor do not cause dynamic instability. Where applicable, a PSS (power system stabiliser) shall be included in the excitation system to ensure proper tuning of the excitation system for stability purposes.

	Symbol	Units
Excitation system type (AC or DC)		Text
Excitation feeding arrangement (solid or shunt)		Text
Excitation system filter time constant	T_f	Sec
Excitation system lead time constant	T_e	Sec
Excitation system lag time constant	T_b	Sec
Excitation system controller gain	K_s	
Excitation system controller lag time constant	T_s	Sec
Excitation system maximum controller output	V_{max}	p.u.
Excitation system minimum controller output	V_{min}	p.u.
Excitation system regulation factor	K_r	
Excitation system rate feedback gain	K_f	
Excitation system rate feedback time constant	T_f	Sec

(f) *Speed governor system, turbine and boiler models*

The generator shall supply a Laplace domain control block diagram in accordance with IEC standard prime mover models for thermal and hydro units (or as otherwise agreed with the TNSP), fully specifying all time constants and gains to fully explain the transfer function for the governor, turbine, penstocks and control systems in relation to frequency deviations and set point operation.

(g) *Control devices and protection relays*

The generator should supply any additional Laplace domain control diagrams for any outstanding control devices (including power system stabilisers) or special protection relays in the unit that automatically impinge on its operating characteristics within 30 seconds following a system disturbance and that have a minimum time constant of at least 0.02 seconds.

(h) *Unit step-up transformer*

	Symbol	Units
Number of windings		
Vector group		
Rated current of each winding		Amps
Transformer rating		MVA _{rated}
Transformer nominal LV voltage		kV
Transformer nominal HV voltage		kV
Tapped winding		
Transformer ratio at all transformer taps		
Transformer impedance at all taps (For three winding transformers the HV/LV1, HV/LV2 and LV1/LV2 impedances together with associated buses shall be provided)		% on rating MVA _{rated}
Transformer zero sequence impedance at nominal tap	Z_0	Ohm
Earthing arrangement, including neutral earthing resistance and reactance		
Core construction (number of limbs, shell or core type)		
Open circuit characteristic		Graph

(j) *Unit forecast data*

The generator shall provide the TNSP and SO with expected maintenance requirements, in weeks per annum, for each unit at a power station.

(k) *Return to service of mothballed generating plant:*

Once the customer has decided to return mothballed generating plant to service, the TNSP requires the information specified for new connections.

(l) *Decommissioning of generating plant:*

Decommissioning of plant is the permanent withdrawal from service of generating plant. The TNSP requires the following with a one-year notice period:

Generator name	
Power station name	
Unit number	
Date to be removed from commercial service	
Auxiliary supplies required for dismantling and demolition	kVA, point at which supply is require, duration

Type of plant	Plant details		Plant details		Plant details
	Plant name	Plant number	Plant name	Plant number	
Renewable					
Non-renewable					
Other					

APPENDIX 16: GENERATOR MAINTENANCE DATA

- (a) The 52-weeks-ahead outage plan per week per generator shall be updated weekly to the SO in the format shown below:

Annual Maintenance Schedule

Station Name: _____

Unit No.	Activity	MW lost	Week										
			1	2	3	4	5	6	7	8	9	10	n+51
1	Annual Maintenance	100											
2	Annual Maintenance	100											
2	Governor modernization	100											
3	Transformer oil	150											
ETC.													

- (b) The annual maintenance/outage plan per generator, looking 15 years ahead, shall also be supplied to the SO. The format shall be quarterly based as follows:

Five Year Ahead Maintenance Schedule

Station Name: _____

Unit No.	Activity	MW lost	Years												
			1	2	3	4	5	6	7	8	9	10	11	12	13
1	Annual Maintenance														
2	Annual Maintenance														
2	Governor modernization														
3	Transformer oil														
etc															

- (c) A weekly variance report, explaining the differences between the planned and actual outages, shall be supplied to the SO as shown below:

Unit No	Outage	Start Date		End Date		Reason for difference
				Plan	Actual	
	Description	Plan	Actual	Plan	ACTUAL	

APPENDIX 17: INFORMATION REQUIREMENTS FOR UPGRADING OF EXISTING CONNECTIONS AND NEW CONNECTIONS

Commissioning	Projected or target commissioning test date
Operating	Target operational or on-line date
Reliability of connection requested	Number of connecting circuits e.g. one or two feeders, or firm/non-firm supply required (subject to Network chapter requirements)
Location map.	Upgrades: Name of existing point of supply to be upgraded and supply voltage. New connections: Provide a 1:50 000 or other agreed scale location map, with the location of the facility clearly marked. In addition, specify the co-ordinates of the point of connection.
Site plan	Provide a plan of the site (with at least 1:200 or 1:500) of the proposed facility, with the proposed point of supply, and where applicable, the transmission line route from the facility boundary to the point of supply, clearly marked.
Electrical single-line diagram	Provide an electrical single-line diagram, drawn to the relevant IEC standard, of the customer intake substation.

APPENDIX 18: GENERATOR HV YARD INFORMATION

The TNSP shall provide the following information to Generators about equipment and systems installed in generator HV yards. The TNSP and the SO shall provide the stability criteria.

Equipment	Requirement
Circuit breaker	MCR rating, peak rating, operating time, OEM, installation date
CT and VT	CT and VT ratings, classes of equipment, burdening, OEM, installation date
Surge arrestor	OEM, age, installation date, number of operations
Protection	Description of protection philosophy for all protection schemes and functions installed, including ARC. Protection reliability information shall be available annually
Power consumption	List of power consumption requirement by equipment requiring supply from AC, DC and UPS
Link	MCR rating, peak rating, OEM, installation date
Outgoing Feeder	MCR rating, peak rating, erection date, length, impedance, transposition characteristics, thermal limits, installed protection, shielding.
Transformer	Transformer specifications for coupling transformers in HV yards. The records of coupling transformers in HV yards must be available on request.
Compressed air system	Compressed air system specifications including schematic drawings
Fault recorder	Fault recorder specifications including resolution, record time, triggering criteria, data format shall be provided on request. The TNSP shall review the fault levels and impedance to network centre from HV yard annually.

APPENDIX 19: OPERATIONAL DATA

This appendix specifies the data format to be used by the SCADA system for the mapping of RTU data into the SCADA database. The database has a definition for each electrical configuration (ELC) or electrical object in the station. Each ELC definition specifies a different ELC type, e.g. transformers, units, feeders, etc, and is accompanied by a picture showing the ELC and all its associated devices as they would be indicated on the SO operational one-line displays. In each instance, the picture defines the primary devices and is followed by the points belonging to each device.

DESCRIPTION OF TABLE COLUMN HEADINGS USED IN THIS SECTION

The headings in the tables of this section are described as follows:

(a) *Device* Gives the name of the device and acts as a collector of all point information belonging to the device.

Each binary status point can be mapped to one or two binary bits. In the case of a breaker or SO lator, the state is reported via two bits. In the case of single-bit alarm points, only one bit is used to report the state of the indication.

01_state This is the alarm state of the point.

10_state This is the normal state of the point.

In the table above the TYPE column indicates the number of bits used to report the state of the point in question.

For single-bit points please ignore the left-hand 0 or 1 value in the headings "01-State" and "10-State".

Where an indication uses two bits to report the state, the right-hand bit is used report that the state is OPEN and the left-hand bit to report the state when it is CLOSED. Thus an open condition will be "01" and a closed state will be "10".

It is thus illogical for a device to have a permanent value of either "00" or "11". However, if the device is in transit between "01" and "10" then a temporary value of "00" is possible. The SCADA system reports a state of "00" as "In transit", which will normally only be seen on slow-moving devices such as Isolators.

(b) *Category* defines the category the point belongs to: Health, Main Protection, Back-up Protection or Information.

Classical alarm systems attempt to set priorities on alarm points. However, the priority of a point changes as the system changes, which means having a fixed priority is not useful. As an alternative, the approach used here is to assign the point to the area that is affected by the indication. In this case we have four areas, namely:

Health	All alarm indications that refer to the health of the primary or secondary plant are assigned to this category.
Main Protection	All protection activity that is triggered by the Main 1 protection circuits is assigned to this category.
Backup Protection	Where back-up protection is installed, such as on transformers, or where Main 2 protection is used, these alarms are assigned to this category.
Information	Pure state change data such the state of a breaker or Isolator are assigned to this category. As such, no alarming is associated with these points – the data presented is pure information.

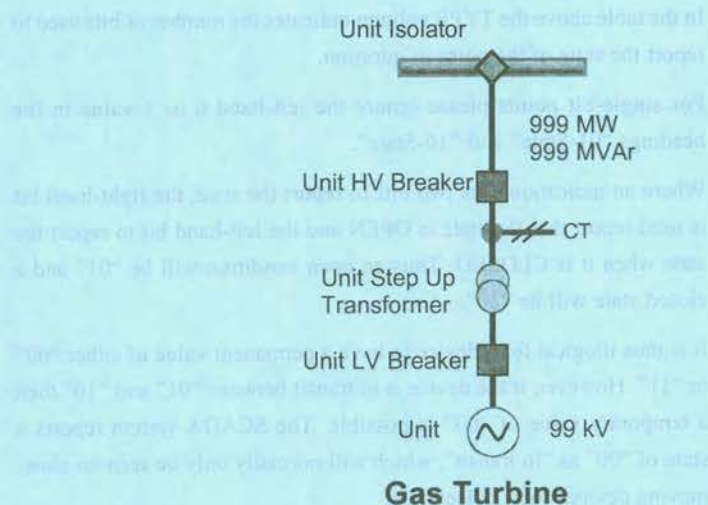
(c) **Type** Indicates the type of point – single-bit, double-bit, analogue or binary change detection.

(d) **Control** Indicates if there is a supervisory control associated with the point

A5.1 Generator

The generator shall install operational measurements to specification from the SO so as to provide continuous operational information for both real time and recording purposes in relation to each unit at each power station in respect of the following:

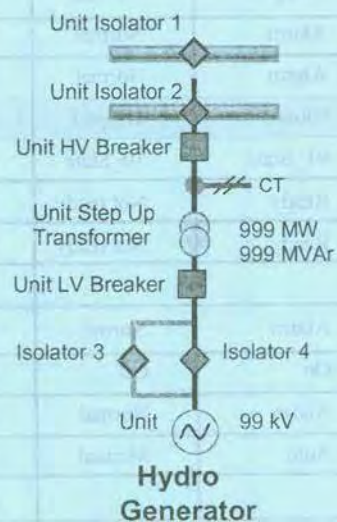
(a) Gas turbines



Unit isolator	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit isolator state	Open	Closed	Info	Double	False
Unit HV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Bus zone trip	Alarm	Normal	Info	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Protection abnormal	Alarm	Normal	Health	Single	False
Protection operated	Alarm	Normal	Main	Single	False
Protection unhealthy	Alarm	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
CT	01_State	10_State	Category	Type	Control
SF6 gas critical (CT)	Alarm	Normal	Health	Single	False
SF6 non-critical (CT)	Alarm	Normal	Health	Single	False
Unit Step Up Transformer	01_State	10_State	Category	Type	Control
Reactive power			Info	Analogue	False
Active power			Info	Analogue	False
Unit LV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
Unit	01_State	10_State	Category	Type	Control
Engine A	Ready	Not ready	Health	Single	False
Engine B	Ready	Not ready	Health	Single	False
Frequency			Info	Analogue	False
Unit start not ready	Alarm	Normal	Health	Single	False
Remote control	On	Off	Info	Single	True
SCO start not ready	Alarm	Normal	Health	Single	False
Sequence starting	Auto	Manual	Info	Single	False
Stator voltage			Info	Analogue	False

Supervisory	Isolated	Off	Info	Single	False
Under-frequency start	Armed	Off	Health	Single	False
Unit at standstill	Yes	Normal	Info	Single	True
Unit auto load to base	Yes	No	Info	Single	True
Unit auto load to minimum	Yes	No	Info	Single	True
Unit auto load to peak	Yes	No	Info	Single	True
Unit failed to start	Alarm	Normal	Health	Single	False
Unit voltage			Info	Analogue	False
Unit in generation mode	Yes	No	Info	Single	True
Unit in SCO mode	Yes	No	Info	Single	True
Unit load rate	Fast	Slow	Info	Single	True
Unit loading mode	Auto	Manual	Info	Single	False
Unit reactive power generated			Info	Analogue	False
Unit reactive power sent out			Info	Analogue	False
Unit active power generated			Info	Analogue	False
Unit active power sent out			Info	Analogue	False
Unit tripped and locked out	Alarm	Normal	Info	Single	False
Unit under frequency start	Initiate	No	Info	Single	False

(b) Hydro units

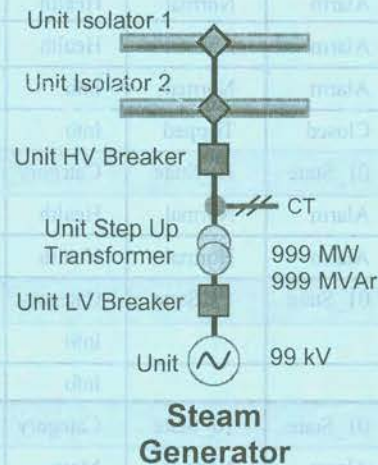


Unit Isolator 1	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit Isolator state	Open	Closed	Info	Double	False
Unit Isolator 2	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit Isolator state	Open	Closed	Info	Double	False
Unit HV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Bus zone trip	Alarm	Normal	Info	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Protection abnormal	Alarm	Normal	Health	Single	False
Protection operated	Alarm	Normal	Main	Single	False
Protection unhealthy	Alarm	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
CT	01_State	10_State	Category	Type	Control
SF6 gas critical (CT)	Alarm	Normal	Health	Single	False
SF6 non-critical (CT)	Alarm	Normal	Health	Single	False
Unit LV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
Unit Step Up Transformer	01_State	10_State	Category	Type	Control
Reactive power			Info	Analogue	False
Active power			Info	Analogue	False
Isolator 3	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit Isolator state	Open	Closed	Info	Double	False
Isolator 4	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False

Unit Isolator state	Open	Closed	Info	Double	False
Unit	01_State	10_State	Category	Type	Control
Frequency			Info	Analogue	False
Auto load	Active	Normal	Info	Single	True
Automatic power factor regulator	Operated	Normal	Health	Single	True
Emergency trip	Operated	Normal	Main	Single	True
Unit start not ready	Alarm	Normal	Health	Single	False
Unit to pump mode	Active	Off	Info	Single	True
Unit to SCO mode	Active	Off	Info	Single	True
Pump start not ready	Alarm	Normal	Health	Single	False
Pump to generation mode	Active	Off	Info	Single	True
Pump to SCO mode	Active	Off	Info	Single	True
SCO start not ready	Alarm	Normal	Health	Single	False
SCO to generation mode	Active	Off	Info	Single	True
SCO to pump mode	Active	Normal	Info	Single	True
Sequence starting	Auto	Manual	Info	Single	False
Stator voltage	Info	Analogue	Log only	No	
Under-frequency start	Armed	Off	Health	Single	False
Unit AGC – high limit	High	Normal	Health	Single	False
Unit AGC – high regulating limit	Health	Analogue	Panel	No	
Unit AGC – low limit	Low	Normal	Health	Single	False
Unit AGC – low regulating limit		Health	Analogue	False	
Unit AGC – raise/lower blocking	Off	Normal	Info	Single	False
Unit AGC – ramp rate			Info	Analogue	False
Unit AGC – set-point active power			Info	Analogue	True
Unit AGC – status	On	Off	Info	Single	False
Unit at standstill	Yes	Normal	Info	Single	True
Unit voltage			Info	Analogue	False
Unit guide vane			Info	Analogue	True
Unit in generation mode	Yes	No	Info	Single	True
Unit in pump mode	Yes	No	Info	Single	True

Unit in SCO mode	Yes	No	Info	Single	True
Unit load limiter			Info	Analogue	True
Unit reactive power generated		Info	Analogue	False	
Unit reactive power sent out			Info	Analogue	False
Unit active power generated			Info	Analogue	False
Unit active power sent out			Info	Analogue	False
Unit synchronising	Yes	No	Info	Single	False
Unit turning in generation direction	Yes	No	Info	Single	False
Unit turning in motor direction	Yes	No	Info	Single	False

(c) Steam units



Unit Isolator 1	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit Isolator state	Open	Closed	Info	Double	False
Unit Isolator 2	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Unit Isolator state	Open	Closed	Info	Double	False
Unit HV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Bus zone trip	Alarm	Normal	Info	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Protection abnormal	Alarm	Normal	Health	Single	False
Protection operated	Alarm	Normal	Main	Single	False
Protection unhealthy	Alarm	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
CT	01_State	10_State	Category	Type	Control
SF6 gas critical (CT)	Alarm	Normal	Health	Single	False
SF6 non-critical (CT)	Alarm	Normal	Health	Single	False
Unit Step Up Transformer	01_State	10_State	Category	Type	Control
Reactive power			Info	Analogue	False
Active power			Info	Analogue	False
Unit LV Breaker	01_State	10_State	Category	Type	Control
Breaker failed to trip	Alarm	Normal	Main	Single	False
Earth applied	Alarm	Normal	Health	Single	False
Pole	Disagree	Normal	Health	Single	False
Breaker gas critical	Alarm	Normal	Health	Single	False
Breaker non-urgent	Alarm	Normal	Info	Single	False
Unit breaker state	Closed	Tripped	Info	Double	False
Steam Unit	01_State	10_State	Category	Type	Control
Unit AGC – high limit	High	Normal	Health	Single	False
Unit AGC – high regulating limit		Health	Analogue	False	
Unit AGC – low limit	Low	Normal	Health	Single	False

Unit AGC – low regulating limit		Health	Analogue	False	
Unit AGC – raise/lower blocking	Off	Normal	Info	Single	False
Unit AGC – ramp rate			Info	Analogue	False
Unit AGC – set-point active power		Info	Analogue	True	
Unit AGC – status	On	Off	Info	Single	False
Unit voltage			Info	Analogue	False
Unit islanded	Alarm	No	Health	Single	False
Unit reactive power generated			Info	Analogue	False
Unit reactive power sent out			Info	Analogue	False
Unit active power generated			Info	Analogue	False
Unit active power sent out			Info	Analogue	False

Generators contributing to regulating reserve shall provide indications for emergency generation and maximum generation for rescheduling.

A5.2 DISTRIBUTOR AND END-USE CUSTOMER

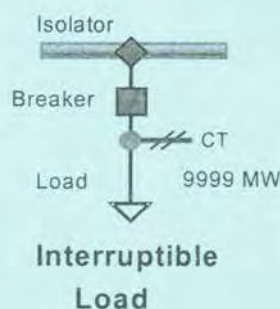
(a) Transmission equipment

The Customer shall provide operational information for both real time and recording purposes in relation to each feeder, transformer and compensation device at each substation required for the full functionality of a SVC, as well as full control by the SO.

Interruptible Load

All interruptible loads shall meet the minimum requirements. The SO shall negotiate and integrate the conditions as presented in bilateral agreements and additional contracts without reducing the requirements as defined in this Grid Code.

The interruptible load shall install operational measurements to specification so as to provide operational information for both real time and recording purposes in relation to each controllable energy block in respect of the following minimum requirements for operation and control of a interruptible load:



Isolator	01_State	10_State	Category	Type	Control
Pole	Disagree	Normal	Health	Single	False
Isolator state	Closed	Open	Info	Double	False
Breaker	01_State	10_State	Category	Type	Control
Unit breaker state	Closed	Tripped	Info	Double	True
CT	01_State	10_State	Category	Type	Control
SF6 gas critical (CT)	Alarm	Normal	Health	Single	False
SF6 non-critical (CT)	Alarm	Normal	Health	Single	False
Load	01_State	10_State	Category	Type	Control
Load reduction acknowledged	No	Yes	Info	Single	True
Load interrupt acknowledged	No	Yes	Info	Single	True
Block load reduction acknowledged	No	Yes	Info	Single	True
Return to service acknowledged	No	Yes	Info	Single	True
Load active power			Info	Analogue	False



APPENDIX 20: POST-DISPATCH INFORMATION

The SO shall provide the following minimum operational information in near real-time and as historic data in relation to each power station:

No	Data description	Format	Size	Unit
1	High limit	Real	999,99	MW
2	Low limit	Real	999,99	MW
3	AGC mode CER/BLO	Character	3	
4	AGC status AUT/OFF/MAN	Character	3	
5	Set point	Real	99,99	MW
6	AGC pulse	Real	9,9	
7	Sent-out	Real	999,99	MW
8	Auxiliary	Real	999,99	MW
9	Contracted Power	Integer	999	MW
10	Station spinning	Integer	999	
11	32-bit flag on AGC settings	Integer		32 bits

The SO shall provide the following minimum operational information in near real-time in relation to the overall dispatch performance:

No	Data description	Format	Size	Unit
1	Area control error (ACE)	Real	999,99	MW
2	Average ACE previous hour	Real	999,99	MW
3	System frequency (Hz)	Real	99,999	MW
4	Frequency distribution current hour	Real	999,99	MW
5	Frequency distribution previous hour	Real	999,99	MW
6	System total generation	Integer	99999	MW
7	Control area total actual interchange	Integer	99999	MW
8	Control area total scheduled interchange	Integer	99999	MW
9	System operating reserve	Integer	99999	MW
10	System sent-out	Integer	99999	MW
11	System spinning reserve	Integer	99999	MW
12	AGC regulating up	Integer	99999	MW
13	AGC regulating down	Integer	99999	MW
14	AGC regulating up assist	Integer	99999	MW
15	AGC regulating down assist	Integer	99999	MW
16	AGC regulating up emergency	Integer	99999	MW
17	AGC regulating down emergency	Integer	99999	MW
18	AGC mode	Char	TLBC /CFC	
19	AGC status	Char	ON/ OFF	
20	Area control error output	Real	999,99	MW
21	System transmission losses	Real	999,99	MW
22	ZESA tie-lines	Integer	99999	MW
23	DRC tie-lines	Integer	99999	MW
24	BPC radial-line	Integer	99999	MW
25	AGC performance indicators			

APPENDIX 21: BACK UP FILES

File	Description	Trigger Event	Frequency
AGC pulses	The total pulses sent to a unit by the AGC system to move the set-point up or down	Ongoing, file created at end of hour	Hourly
NERC	A list of the total NERC "A1" and "A2" criteria violations in the system over an hour	Ongoing, file appended at end of hour	Daily
System near real-time data	Historic near real-time system data files on readings as required for post-dispatch	Communication failure	To be agreed
Unit near real-time data	Historic near real-time unit data files on readings as required for post-dispatch	Communication failure	To be agreed

The AGC will provide the following near real-time information in near real-time or within 10 minutes of the event dispatch performance:

File	Description	Frequency	Unit
1	AGC control error (ACE)	Real	MW
2	AGC set-point error (ASPE)	Real	MW
3	System frequency (Hz)	Real	MW
4	Frequency distribution current flow	Real	MW
5	Frequency distribution previous hour	Real	MW
6	System total generation	Integer	MW
7	Control area total actual interchange	Integer	MW
8	Control area total scheduled interchange	Integer	MW
9	System operating reserve	Integer	MW
10	System set-point	Integer	MW
11	System spinning reserve	Integer	MW
12	AGC regulating up	Integer	MW
13	AGC regulating down	Integer	MW
14	AGC regulating up limit	Integer	MW
15	AGC regulating down limit	Integer	MW
16	AGC regulating on emergency	Integer	MW
17	AGC regulating down emergency	Integer	MW
18	AGC mode	Char	TRBC, CTC
19	AGC status	Char	ON, OFF
20	Area control error output	Real	MW
21	System transmission losses	Real	MW
22	AGC set-point	Integer	MW
23	AGC set-point	Integer	MW
24	AGC set-point	Integer	MW

APPENDIX 22: GENERATOR PERFORMANCE DATA

Generator performance data is a means of providing a measurement of generating station processes. A combination of different metrics yields indicators that provide an objective assessment of critical parameters on the performance of key processes. They are used to identify non-performing areas in order to facilitate improvements. This is also used to provide industry with statistical data on how well the industry is performing.

The indicators will provide generating stations in Zambia an opportunity to benchmark their performance against other best run hydro power stations in the world. It also offers these station's management data for strategic management.

The Key Performance Indicators report on each station's efficiency and effectiveness is at a tactical level. The assumption here is that, even when a generating station is embedded in a vertically integrated utility, it is very possible to measure these statistics against a common benchmark.

For the purpose of these indicators, all generation equipment outages will be considered planned and scheduled at every beginning of the year. All other outages procured during the progression of the year will be considered unplanned.

(a) Reliability

This indicator will be expressed in terms of Unit Capability Factor (UCF). The purpose of this indicator is to monitor progress in attaining high unit and plant energy production availability. It reflects the effectiveness of plant programmes and practices in maximizing available electrical generation and provides an overall indication of how well the plant was operated and maintained. It comprises Planned Capability Loss Factor (PCLF) and Unplanned Capability Loss Factor (UCLF) and it is expressed as a difference between 100% and the percentage attributed to PCLF and UCLF.

$$UCF = 100 - PCLF - UCLF$$

(i) Unplanned capability loss factor (UCLF)

The purpose of this indicator is to monitor industry progress in minimising outage time and power reductions that result from unplanned equipment failures or other conditions. This indicator reflects the effectiveness of plant programmes and practices in maintaining systems available for safe electrical generation.

It is therefore an expression of energy lost due to unplanned outages, against the total available energy, calculated as a percentage.

$$UCLF = \frac{\text{UnplannedMWhlost} * 100\%}{\text{TotalMWhde signed}}$$

(ii) *Planned capability loss factor (PCLF)*

Planned capability loss factor is defined as the ratio of the planned energy losses during a given period of time to the reference energy generation expressed as a percentage.

Planned energy loss is energy that was not produced during the period because of planned shutdowns or load reductions due to causes under plant management control. Energy losses are considered to be planned if they are scheduled and appear on the System Yearly Maintenance Program. It will stimulate detailed scrutiny and predictability of generating plant so that outages are focused and scheduled one (1) year in advance.

$$PCLF = \frac{\text{Planned MWh lost} * 100\%}{\text{Total MWh designed}}$$

(b) *Internal Reliability Loss Factor*

This factor is expressed to measure the number of internal factors leading to reduction in Reliability of generating plant against the total reduction in Reliability. It is therefore expressed as a percentage of energy lost due to internal faults leading to lost generation against available energy during the period.

$$IRLF = \frac{\text{Unplanned MWh lost due to internal faults} * 100\%}{\text{Total MWh available}}$$

(c) *External Reliability Loss Factor*

This factor measures the influence that external factors leading to forced outages has on the generating plant. Such factors could include transmission line trippings or outages, deferred maintenance activities due to system load changes, cascade trippings and many others. It is expressed as a percentage of energy losses due to faults external to the generating plant, leading to loss of generation against total available energy during the given period.

$$ERLF = \frac{\text{Unplanned MWh lost due to external faults} * 100\%}{\text{Total MWh Available}}$$

(d) *Mean-Time-To-Repair (MTTR)*

This factor measures the average time used to carry out repairs due to faults in any given period. It is expressed as the sum of all durations generating units were disconnected from the grid due to internal faults against the total number of breakdowns due to internal faults.

$$\frac{\sum_{n=1}^n I_n}{T} * 100\%$$

Where:

t = time duration for each internal fault

n = *outage* number

T is the total number of *outages* in the period.

(e) *Mean-Time-Between-Failures (MTBF)*

This is expressed as the total time in a given period divided by the number of internal faults plus 1. It is meant to express on an average the time between any two failures following each other.

Mean-Time-To-Repair and Mean-Time-Between-Failures are two factors that are used to measure the quality of maintenance being carried out on generating plant.

(f) *Start-up Failure Rate (SFR)*

This is expressed as the number of times a generating unit or plant fails to connect to the grid at the required time against the total number of requested synchronizations. This measure is used to gauge under-frequency recovery time.

$$SFR = \frac{\text{Number Of failed Start Ups} \times 100\%}{\text{Number Of Contracted Start Ups}}$$

Start-up comprises the set of operations that enable the unit to be connected to the off-site power grid for the production of electrical energy according to requirements issued by the SO. It includes all the necessary “de-isolations” and rendering to “normal” all equipment required for generation.

The contracted start-up refers to the number of start-up instructions issued and received from the System Operation by the Generation Control Room.

(g) *Total Unplanned Station Shutdowns (TUSS)*

This is the total number of times that a generating station separates with the grid. It could either be as a result of an internal or external unplanned factor, but leading to a total station outage. It measures the protection requirements defined in the Network chapter, section 4.1, or it could be declared as an Ancillary Service as declared in the System Operations chapter section 7.

APPENDIX 23: REPORTING FORMAT FOR PERIODIC TESTING OF UNDER-FREQUENCY LOAD SHEDDING RELAYS

Distributor:

Date:

Substation:

Fed from transmission substation (directly or indirectly):

	Activating frequency		Timer setting	
	Required	As tested	Required	As tested
Stage 1	48.8Hz		Instantaneous	
Stage 2	48.5Hz		Instantaneous	
Stage 3	48.2Hz		Instantaneous	
Stage 4	49.0Hz		20 seconds	
Stage 5	0.8Hz/s		Instantaneous	
	Feeders selected (Required)		Feeders selected (As tested)	
Stage 1				
Stage 2				
Stage 3				
Stage 4				

(Current SAPP Guidelines)

APPENDIX 24: PERFORMANCE INDICATORS FOR TNSPS AND SO.

Indicator	Month	Year to date	12 MMI	Unit
System minutes lost				Minutes
No. of Interruptions				
No. of statutory voltage transgressions				
Mandatory under-frequency load shedding				

APPENDIX 25: INFORMATION CONFIDENTIALITY

CONFIDENTIALITY AGREEMENT FOR INFORMATION TRANSFER TO THIRD
PARTIES

CONFIDENTIALITY AGREEMENT

BETWEEN

(HEREINAFTER REFERRED TO AS THE INFORMATION OWNER)

AND

(HEREINAFTER REFERRED TO AS THE RECIPIENT)

IN RESPECT OF INFORMATION SUPPLIED TO PERFORM THE FOLLOWING WORK:

(HEREINAFTER REFERRED TO AS THE WORK)

ON BEHALF OF

(HEREINAFTER REFERRED TO AS THE CLIENT).

1. The Recipient agrees to treat all information (hereinafter referred to as the Information) received from the Information Owner, whether in hard copy or electronic format or in any format whatsoever, as strictly confidential.
2. The Recipient agrees to disclose the Information only to authorized persons who are in his permanent employ, and who require access to the Information to perform their duties in respect of the Work on behalf of the Client.
3. Persons other than those described in Clause 2 above, including but not restricted to temporary employees, subcontractors, and sub-consultants, shall enter into separate Confidentiality Agreements with the Information Owner prior to receiving the Information.
4. The Recipient undertakes to use the Information only to perform the Work on behalf of the Client, and for no other purpose whatsoever.
5. On completion of the Work, the Recipient shall at his expense return to the Information Owner all hard copy material and electronic media containing the Information supplied to him by the Information Owner. The Recipient shall furthermore ensure that all duplicate copies of the Information in his or his employees' possession (electronic as well as hard copy format) are destroyed.
6. The Recipient shall take all reasonable measures to protect the security and integrity of the Information.

7. If requested to do so by the Information Owner, the Recipient shall forthwith at his expense return to the Information Owner all hard copy material and electronic media containing the Information supplied to him by the Information Owner. The Recipient shall furthermore ensure that all duplicate copies of the Information in his or his employees' possession (electronic as well as hard copy format) are destroyed.
8. The Recipient shall report any leak of the Information, howsoever caused, to the Information Owner as soon as practicable after he becomes aware of the leak, and shall provide to the Information Owner with all reasonable assistance to ensure its recovery or destruction (as deemed appropriate by the Information Owner).

Signed at on this the day of
 by (full name) in his/her
 capacity as on behalf of, the
 Information Owner.....

Signed at on this the day of
 by (full name) in his/her
 capacity as on behalf of, the
 Recipient

APPENDIX 26: PLANNING SCHEDULES

Schedule 1: Five and Fifteen-year demand forecast

	<i>Demand = Total Demand + Distribution Losses – Embedded Generation</i>				
	<i>Energy</i>	<i>Maximum demand</i>		<i>Expected minimum demand</i>	
<i>Year</i>	<i>GWh</i>	<i>MW</i>	<i>MVar</i>	<i>MW</i>	<i>MVar</i>
Measured					
Year 0					
Year 1					
Year 2					
Year 3					
Year 4					
Year 5					
Year 6					
Year 7					
Year 8					
Year 9					
Year 10					
Year 11					
Year 12					
Year 13					
Year 14					
Year 15					

